



FAI Sporting Code

Section 4 – Aeromodelling

Volume EDIC

Electronic Devices in Competition

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Ver. 1.0, 2026-01-01

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This volume, instituted in January 2014, will contain various special technical specifications, guidance and approval procedures relating to specific electronic devices regardless of the class in which those devices may be utilized.

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The use of “shall” and “must” implies that the aspect concerned is mandatory. The use of “should” implies a non-mandatory recommendation; “may” indicates what is permitted, and “will” indicates what is going to happen. Words of masculine gender shall be taken as including the feminine gender unless the context indicates otherwise. Words expressing the singular will be taken to include the plural and vice versa. Italics are used for explanatory notes.

Revisions of 2025 code

Paragraph	Date of Publication	Brief description of change	Change incorporated by
Section 2	January 2026	Section 2 - SM	Trip Barber Chris Flanigan
Section 7	January 2026	New Section 7 - F5L	Manfred Lex EDIC chairman

The following amendments are also shown for reference

Amendments for Reference			
Paragraph	Date of Publication	Brief description of change	Change incorporated by
Section 1	January 2025	2.4 (c) added to note a clarification of motor on command	Manfred Lex EDIC chairman
Section 6	January 2025	Added 4.1 k less than 100W below 1.2ms input signal	Harald Viehweger EDIC F3G
Section 6	February 2024	Aligned overall structure of Section 6 (F3G) according to the previous sections of the document Added specification and clarification regarding "Motor Run Timer" and "Start Height" Extended paragraph 1 - 5 of this section to improve consistency and transparency.	Harald Viehweger Manfred Lex EDIC
Section 3	January 2023	2.2b alternate negative battery lead 2.2d alternate post motor stop control strategy 2.3.1 new voltage, current, temperature limits for 2% accuracy 2.4d add accumulation start with currents > 5A	Ian Kaynes F1 subcommittee
Section 4	February 2022	added F5B Limiter/Logger requirements	Manfred Lex
Section 3	January 2022	update with additional requirements from lessons learned	Manfred Lex
Section 6	January 2022	added section 6: F3G	Manfred Lex
Section 1	June 1 2020	Amended Section 3 chapter 2.2f: CAT2 events (World Cup, Open International) allow restart of motor with setting height to "---" and hence scoring of zero. 2.3.10 Firmware revision shown in display 3.4.5 Motor respond allowed for CAT2 events 5.2.2.j testing procedure motor restart	Manfred Lex EDIC chairman
Section 5	June 1 2020	added section 5: Free Flight devices and rules	Ian Kaynes F1 subcommittee
Section 3	January 1 2020	Amended Section 3: F5D Energy Limiter V 1.0 throughout, as a consequence of the change from F5D to F3E. Only designators have been changed e.g. F5D → F3E and F5DEL → F3EEL.	Kevin Dodd Technical Secretary
Section 3	July 1 2016	Added Section 3: F5D Energy Limiter V 1.0	Kevin Dodd Technical Secretary
Section 4	July 1 2016	Added Section 4: Performance Requirements Definition for FAI-F5B – Real Time Energy Consumption Telemetry (REAL-TECT) System – compatible with existing 'GASSENOR' equipment.	
Page 5	2016	Consequential changes for Volume ABR, due to the release of Edition 2016 of FAI General Section.	Kevin Dodd Technical Secretary
Section 2	2016	Added Section 2: Technical guidance and specification for altimeters used in Space Modelling competition.	
Appendix A	n/a	Deleted, since the list of Approved AMRTs/Firmware is updated as necessary, and is available in the Documents section of the CIAM website under the heading EDIC WG & Electronic Device Approvals	

Paragraph	Date of Publication	Brief description of change	Change incorporated by
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There were no changes or additions made at the 2014 Plenary meeting.

Paragraph	Date of Publication	Brief description of change	Change incorporated by
Section 1	2014	Added the first section: technical specification and guidance for F5J altimeter/motor run timer (AMRT).	Jo Halman Technical Secretary
Addendum		Added the EDIC Working Group Terms of Reference	

RULE FREEZE FOR THIS VOLUME

This content of this volume is not subject to Plenary meeting approval, nor is it restricted by any rule freeze regulation. It is under the direct control of CIAM Bureau on recommendation from the EDIC Working Group and may be updated at any time during the year.

Section 1 - F5J Altimeter/Motor Run Timers V 1.1

Initial Publication: 01 January 2014

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Appendix A has been deleted (2015)

The List of Approved AMRT Firmware is available in the Documents section of the CIAM website under the heading EDIC WG & Electronic Device Approvals.

1.0 INTRODUCTION

1.1 Purpose of this Document

The FAI CIAM rule making/amendment process is geared to maintain competition rule stability and as such presents some difficulties in areas subjected to rapidly changing technologies. To overcome this problem, CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” Working Group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of Altimeter/Motor Run Timers (AMRTs) for use in F5J competitions. The information contained herein is relevant to AMRT manufacturers, competition organisers, competitors and persons involved in the evaluation of AMRTs.

1.2 Philosophy behind the AMRT Specification

International competitions attract large numbers of competitors, each of whom is entitled to three models. Consequently, competition organisers could expect to be presented with a significant number of AMRTs from a number of different manufacturers for scrutineering and verification purposes.

It is against this background that the AMRT Specification has been formulated with the following objectives:

- a) To accurately determine the ‘Start Height’ as required for the competition.
- b) To minimise any ambiguity in reading the ‘Start Height’ from the AMRT.
- c) To provide a safe working environment for official timekeepers when reading the AMRT.
- d) To ensure that the AMRT can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- e) To retain ‘Start Height’ information for subsequent resolution of any disputed reading.
- f) To provide a standard method of operation so that, irrespective of AMRT manufacture, verification testing can be performed easily with a defined test procedure.
- g) To define standard connectors for compatibility with test facilities and to reduce the risk of accidental damage during handling by competition officials.
- h) To provide a standard method of determining that the AMRT is of the type for which an approval has been granted.

2.0 INFORMATION FOR MANUFACTURERS

2.1 AMRT Technical Specification

This section is the formal specification for AMRTs intended for use in the F5J competition environment.

2.2 An electronic AMRT carried in an F5J model shall fulfil the following technical specifications:

- a) It must use the barometric pressure measurement technique.
- b) Altitude indication must be based on the International Standard Atmosphere, as defined in ICAO Document 7488/2.
- c) The method of converting pressure measurements to height information must be such that the specified accuracy is maintained across the full working range of the AMRT.
- d) It must record the maximum difference in pressure altitude (‘Start Height’) in accordance with the following definition:

‘Start Height’ is defined as the recorded maximum difference in pressure altitude, measured in metres, from initialisation until 10 seconds after the motor is stopped manually by the competitor or automatically by the Motor Run Timer, whichever occurs first.
- e) The Motor Run Timer must stop the motor 30 seconds after it is started at launch, if it has not already been stopped by the competitor within this time limit.
- f) The AMRT must not allow any restart of the motor at any time during the flight except for CAT2 (World Cup, Open International) events. Consequently, the start height has to be set to not recorded “---”.

- g) The AMRT must have compatibility with ESCs using the nominal 1 to 2 milliseconds control signal from the receiver such that no low-speed motor drive occurs when “motor stop” is instituted either by the competitor or the AMRT.
- h) The supply of power to the AMRT shall be that available from the receiver output to the ESC at the receiver motor command connector. If the power source to the receiver is supplied via the ESC motor command connector, it is permissible for this to be supplied to the receiver via the AMRT.
- i) The connectors of the AMRT shall be of a universal type compatible with JR/Futaba and must be proof against any misalignment that could result in electrical damage to the AMRT or connected equipment. Where an extension lead is used to satisfy this requirement, it must be secured to the AMRT in such a manner that it cannot be unintentionally removed.
- j) The AMRT shall be provided with a digital display. The display may be an integral part of the AMRT, or a separate module that is either carried within the model or is detachable.
A detachable AMRT display may be used to service any number of AMRTs of the same approved type.

2.3 Characteristics of the AMRT

- 2.3.1 The accuracy of the determined ‘Start Height’ shall be within the range plus or minus 2.5 metres. This accuracy must be achieved across an ambient ground level pressure range of 750 to 1050 hPa and across an ambient temperature range of 0 to +50 degrees centigrade.
- 2.3.2 The calculation of ‘Start Height’ shall incorporate any calibration parameters provided by the pressure sensor manufacturer.
- 2.3.3 The sampling rate of the pressure measurement used for ‘Start Height’ determination shall be a minimum of 10 samples per second.
- 2.3.4 Any firmware processing of pressure measurements (or height information derived from such measurements) used for ‘Start Height’ determination, must not result in a modification of the maximum height as would be detected by the single measurement sample representing the peak value.
- 2.3.5 The timing accuracy used for the Motor Stop function and ‘Start Height’ determination shall be equal to or better than 0.1 seconds.
- 2.3.6 There must be no configurable settings or adjustments that allow the user to modify the operation of the AMRT such that it will operate outside the requirements of the competition rules.
- 2.3.7 Any additional features incorporated into the AMRT by the manufacturer, must not cause any variation in the sampling rate or timing functions, nor affect the ability of the AMRT to satisfy the requirements of the competition rules.
- 2.3.8 The resolution of the display when showing the calculated ‘Start Height’, shall be 0.1 metre, this to be displayed with the format “xxx.x”. Leading zero characters may be blanked. (When using the final calculated ‘Start Height’ for scoring purposes, the reading shall be rounded down to the nearest full metre.)
- 2.3.9 On completion of the period ending 10 seconds after “motor stop”, the recorded ‘Start Height’ shall be continuously transferred to the display with a minimum repetition rate of once per two seconds. This function shall commence automatically when the recording of the ‘Start Height’ has been completed.
- 2.3.10 The display shall show the firmware revision level of the AMRT, in the format “Fx.x” (for first category events) or “Sx.x” (for secondary category events) with any leading zero characters blanked.
- 2.3.11 Display of the firmware revision level shall occur for the first 3 seconds following the application of power to the AMRT. On completion of the 3 seconds period, the display shall continue to show any previously recorded ‘Start Height’, or cleared status indication, until such time as a Motor Start signal is detected. This sequence shall occur irrespective of whether an input command signal is being applied to the AMRT.
- 2.3.12 If the display is not an integral part of the AMRT, the method of transferring the information to the display shall be immune to any form of corruption or modification from external sources.

2.4 Recording the ‘Start Height’

The AMRT shall record the ‘Start Height’ in accordance with the following procedure:

- (a) Immediately on application of power to the AMRT it must determine a zero height reference level from which any subsequent ‘Start Height’ will be determined. Any previously recorded ‘Start Height’ must NOT be erased at this point.

- (b) This function shall be carried out with the model at ground level at the location specified within the competition rules.
- (c) A maximum of 5 seconds is permitted for the zero-height reference level to be determined, during which period the AMRT shall not respond to any Motor Start signal instituted by the competitor or resulting from receiver start up procedures.

Note: A Motor Start command is defined as any control signal exceeding 1.2 milliseconds. Before the motor start command the AMRT shall not output a control signal greater than 0.9 milliseconds which means motor-off.

- (d) After power has been applied to the AMRT, the 'Start Height' determination and the start of the Motor Run Timer will be initiated by the Motor Start signal instituted by the competitor. Such initialisation must clear any previously recorded 'Start Height'.
- (e) The cleared status of any previously recorded value shall be indicated on the display as a sequence of dashes (-----). This shall be carried out in such a manner that if a detachable display were to be connected it would show the cleared status.
- (f) When the Motor Start signal has been instituted by the competitor, the AMRT will determine, and record, a new 'Start Height' replacing the previously cleared status.
- (g) When power is removed from the AMRT, the 'Start Height' or cleared status indication must be retained by the AMRT. The application of power to the AMRT must never erase either the recorded 'Start Height' or the cleared status indication. Only the Motor Start signal instituted by the competitor may do this.

3.0 TECHNICAL CONSIDERATIONS

This section is provided as design guidance on AMRTs intended for use in the F5J competition environment.

3.1 Restriction on User Configuration Options

Whilst for commercial considerations manufacturers may wish to produce AMRTs adaptable for differing applications, when used in the F5J competitive environment, the installed firmware shall have dedicated F5J functionality with no user configurable options. The purpose of this is to eliminate any possibility of errors in operation due to incorrect configuration settings.

3.2 ISA Calculation

Pressure at a given location is caused by the weight of the atmosphere above. This varies with air density, humidity, gravity and temperature. In addition, temperature itself reduces as height increases. Consequently, height is not directly related to pressure and some form of standardized definition is required to enable instrumentation based on barometric pressure to be used to determine height.

The International Standard Atmosphere (ISA) assumes a mean sea level pressure of 1013.25 hPa at a temperature of 15 degrees C, a specified temperature lapse rate with height, a specified gravitational force and a given molecular density of air.

Conditions on any day are unlikely to correspond exactly with the ISA model; a typical spread of 50hPa can occur at a given location due to normal meteorological conditions. Consequently, significant variations will occur from true geographic height. However, for aviation applications, including F5J, all instruments calibrated and operating in accordance with the ISA standard, will produce the same result.

- 3.2.1 In the lower atmosphere (up to 12km), the ISA relationship for a pressure P, can be described by the following formula:

$$\text{Altitude} = K1 \times \{1 - (P/Po)^{K2}\} \text{ metres}$$

$$\text{where } K1 = 44330.76923 \text{ metres}$$

$$K2 = 0.190266669$$

$$Po = 1013.25 \text{ hPa (ISA sea level pressure)}$$

K2 is derived from the universal gas constant, temperature lapse rate, gravitational force, molecular mass of dry air and temperature with respect to absolute zero.

This formula results in a relationship between pressure and height that is represented by a curve that is steeper at high pressure/low altitude and less steep at low pressure/high altitude. Over the range of site elevations that might be encountered for F5J competitions, the nature of the ISA relationship is significantly non-linear. There is approximately 14% difference in the pressure reduction of a 200 metre launch commencing at sea level compared with one starting at 1500 metres.

- 3.2.2 The calculation of F5J "Start Height" requires a precise measurement of the difference between two ISA pressure altitudes. The first is the zero reference value at ground level for the local site elevation. The second is the maximum altitude that occurs during the launch phase of the aircraft.
- 3.2.3 In order to achieve sufficient accuracy for F5J, the method of firmware processing of pressure data is important. Whilst ideally the ISA computation should be performed by an arithmetic calculation with

maximum precision, processing constraints may make this impractical. It is permitted to use an algorithm approach to emulate the computation provided it can be demonstrated that an appropriate accuracy is maintained and no discontinuities are introduced in the ISA curve.

- 3.2.4 In order to maintain overall AMRT accuracy within the specified tolerance, the accuracy of the firmware ISA computation should be within the range plus or minus 0.5 metres.
- 3.2.5 AMRT manufacturers may wish to adopt pressure sensor modules that have an inbuilt pressure-to-height conversion. In such a case, it must be demonstrated that the computation complies with the accuracy requirements over the specified working range of the F5J application.

Note: In this context, results from calculations using the formula of the 1976 US Standard Atmosphere, agree closely with those of the ISA formula in the lower atmosphere and are considered acceptable over the F5J working range.

- 3.2.6 Pressure sensor offset errors will influence the 'Start Height' calculation. However, since 'Start Height' is the difference between two readings, much of any offset error is removed from the result.

3.3 'Start Height' Determination

- 3.3.1 'Start Height' determination requires the AMRT to perform a peak (height) detection function for the time duration commencing at switch-on initialisation, until 10 seconds after the motor has been stopped.
- 3.3.2 The F5J competition application requires the 'Start Height' to be determined with the best possible accuracy and all manufacturers' AMRTs should exhibit similar characteristics.
- 3.3.3 Initialisation following power-up, and the acquisition of the ground level (zero height reference) reading should be accomplished with due allowance for any time required for the pressure sensor device to stabilise.
- 3.3.4 During the launch phase, the dynamics of the flight envelope are subject to rapid change and the transition through the peak can occur within a short time period. Item 2.3.3 defines the minimum acceptable sampling rate for detecting the peak. It is not permissible for additional filtering or smoothing to be incorporated in the firmware as this can result in a downgrading of the ability to accurately detect the peak value.
- 3.3.5 Oversampling functions incorporated within the pressure sensor device may be used subject to being configured in a manner that maintains the integrity of readings taken at the specified sampling rate.
- 3.3.6 Pressure sensor devices:
 - (a) Are subject to variation due to temperature. Many sensor manufacturers specify the inclusion of temperature data in the processing of the pressure data. Manufacturers' recommended firmware procedures must be incorporated into the AMRT firmware.
 - (b) Can be affected by strong light falling on the sensitive area, therefore they must be mounted within the AMRT in such a manner that the sensitive area is shielded from strong light levels.
- 3.3.8 The accuracy tolerance stated in item 2.3.1 is the maximum permitted when all possible variations are at worst case conditions. No individual factor should contribute a significant proportion of the total permitted variation.

3.4 Motor Run Timer

- 3.4.1 The Motor Run Timer function must operate with the accuracy specified in item 2.3.5.
- 3.4.2 On initial power-up of the AMRT, the Motor Run Timer must be protected from false triggering during the period defined in Item 2.4 (c). After this period, the Motor Run Timer shall commence operation and 'Start Height' determination shall be initiated on the first occasion that the command signal exceeds 1.2 milliseconds.
- 3.4.3 After the Motor Run Timer has been triggered, the timing of the motor run ceases when either the command signal reduces below 1.2 milliseconds, or the elapsed time reaches 30 seconds and a definite stop command must be applied to the ESC by the AMRT.
- 3.4.4 The 'Start Height' determination process shall continue for a further 10 seconds after the instant of "motor stop".
- 3.4.5 Once the definite stop command has been applied to the ESC, the system must not respond to any further changes in the motor command signal except for CAT 2 events where a motor restart is allowed.
- 3.4.6 A small amount of hysteresis is permitted in the sensing of the 1.2 milliseconds motor command signal to eliminate malfunction with jittering command signals.

- 3.4.7 Monitoring of the timing functions is achieved with the aid of the displayed information as detailed in Items 2.4 (a) – (g).

3.5 Retention of ‘Start Height’ Reading

- 3.5.1 In the normal competition environment, the ‘Start Height’ reading is obtained immediately following the landing of the aircraft, without power removal from the system.
- 3.5.2 There is a requirement for the AMRT to retain the ‘Start Height’ reading when power is removed. The primary reason for this is for purposes of reading dispute resolution and possible AMRT investigation away from the aircraft installation.
- 3.5.3 Erasure of the stored ‘Start Height’ reading shall only occur in accordance with the sequence detailed in Items 2.4 (a) – (g).

3.6 Display Considerations

- 3.6.1 The display readout should employ characters that are easily legible in normal outdoor ambient light levels. It is not acceptable to require adjustment of contrast levels to suit ambient conditions.
- 3.6.2 Character height should be such that no optical aids other than normal spectacles are required in order to observe the readings.
- 3.6.3 The formation of the numeric font must be unambiguous. With some font formats it may be necessary for manufacturers to substitute a capital letter “O” character to replace a slashed through zero numeral in order to avoid confusion with the numeral 8.

3.7 ESC Compatibility

There is no defined standard for the manner in which ESCs (electronic speed controllers) initialise to the range of motor command signals. In addition, programmable transmitters allow the user to set the motor command signals to any value within the programmable range.

- 3.7.1 Commercial pressures will encourage AMRT manufacturers to configure their products to operate with the majority of available ESC devices.
- 3.7.2 The F5J specification of motor on threshold at 1.2 milliseconds has been selected as being compatible with most known equipment but cannot be a guarantee of correct operation for every combination of transmitter, receiver and ESC.

3.8 Connector Compatibility

- 3.8.1 AMRTs exist that do not fully satisfy the shrouded connector requirement. Whilst a bare pin arrangement may be acceptable for many other applications, the F5J competition environment requires event officials to handle competitors’ equipment for verification purposes. There is, therefore, the potential to cause accidental damage. For this reason, the use of shrouded connectors has been specified. It is permissible to supply an extension cable to meet the shrouded connector requirement provided that it is mechanically restrained at the connection to the AMRT in such a manner that it cannot be unintentionally removed.
- 3.8.2 Where the readout display is detachable, the AMRT manufacturer must ensure that where a non polarised connector is used, incorrect insertion cannot cause damage to the AMRT or display.

4.0 COMPETITION INFORMATION

4.1 Installation Environment

- 4.1.1 The installation and functioning of the AMRT, the transmitter and its programming, receiver and ESC, is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.
- 4.1.2 The AMRT must be installed in such a way that it is protected from pressure fluctuations other than changes in atmospheric pressure which result from the height of the model above ground level.
- 4.1.3 The AMRT must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the true barometric pressure at any time is prohibited.
- 4.1.4 Adequate venting shall be provided to minimise any uneven pressure distribution within the model in the region where the AMRT is installed.
- 4.1.5 The ESC must always operate via its series connection to the AMRT and not with a direct connection to the receiver.
- 4.1.6 Where the readout display is an integral part of the AMRT the AMRT shall be installed in such a manner that it is readily accessible for viewing the readings.
- 4.1.7 Where the readout display is not an integral part of the AMRT, the connection facility for the display unit shall be readily accessible for the purpose of connecting the display unit to view the readings.

cont/...

- 4.1.8 The competitor's AMRT shall be installed in such a manner that it can be easily removed from the model for verification purposes at any stage during the period of the contest.
- 4.1.9 The connectors linking the AMRT to the receiver shall be readily accessible so that a monitoring AMRT with appropriate interconnection can be fitted on demand by the Contest Director. Such monitoring AMRTs will serve to verify 'Start Height' whilst maintaining the normal operation of the competitor's own installation.
- 4.1.10 If a competitor incorporates an additional lead extension for use with a detachable display, it is the competitor's responsibility to ensure compliance with section 3.8.2.

5.0 VERIFICATION OF AMRTs

This section details procedures for verifying the operation of Approved AMRTs for use in the F5J competition environment. It is provided as a guide for both competitors and competition organizers.

5.1 Purpose of Verification

All AMRTs used in an F5J competition are required to be of an Approved Type. The verification process serves to ensure that no individual AMRT operates in a manner different from others in use at the competition. As such, comparison methods are employed for 'Start Height' verification, in place of a requirement to undertake absolute accuracy measurements.

5.2 Verification Procedures

5.2.1 Firmware Revision Level

An AMRT with connected display, will show the firmware revision level for three seconds immediately after the application of power.

5.2.2 Operational Sequence

For this procedure, the AMRT has to be supplied with power, a command signal input and a means of monitoring the motor "run/stop" signal that is supplied to the ESC.

A display module must be connected when not built in.

Whilst this procedure is applicable to a complete aircraft installation, for reasons of safety and when verifying an AMRT, it should be removed from the aircraft and tested independently.

The procedure is as follows:

- (a) With the command signal at Motor Off, apply power to the AMRT. After the initial display of the firmware revision level, the display will show either a previously recorded 'Start Height' or a sequence of dashes. The ESC signal will be in the "motor stop" state.
- (b) If a sequence of dashes is shown the first time power is applied, it will be necessary to perform steps (c) and (d) to establish a 'Start Height'
- (c) Apply a Motor On command and raise the AMRT to a low height. After a short time period, apply a Motor Off command. The display will show a sequence of dashes from the time of the Motor On command to 10 seconds after the Motor Off command. During this sequence the ESC signal follows the command signal.
- (d) After the 10 seconds has elapsed, the display will show a low value height reading.
Note: The value displayed may be influenced by local air movement.
- (e) Remove power. Wait for approximately 30 seconds and then, with the Motor Off Command, selected, re-apply power. After the display of the firmware revision level, the same low level height reading as recorded in (d) will be displayed.
- (f) Apply and maintain a Motor On command. Immediately the display will show a sequence of dashes and the ESC signal becomes "motor run".
- (g) 30 seconds after the Motor On command, the ESC signal will change to "motor stop". Immediately raise the AMRT to a low height for approximately two seconds to simulate a peak height.
- (h) 10 seconds after the ESC signal became "motor stop", the display will show a low height value.
- (i) Apply a Motor Off command.
- (j) Whilst maintaining power, apply a further Motor On command. The ESC signal will remain in the "motor stop" condition and the AMRT will remain in the same state. In case the SW is enabled for CAT2 events, the ESC signal will go to "motor run" condition and the display will show "---".

Any further Motor On/Off command will put the ESC signal into “motor run” or “motor stop” condition respectively.

- (k) Repeat steps (e) to (h), but instead of waiting for 30 seconds at step (g), apply a Motor Off command after about 10 seconds which will cause the ESC signal to become “motor stop”. The result of the “motor stop” should be as (h) and (j).

5.2.3 ‘Start Height’ Detection

Verification of the measured ‘Start Height’ value requires the use of a ‘vacuum’ chamber. Such a chamber comprises an airtight container from which air can be extracted to create a pressure reduction. Air equivalent to approximately 3% of the chamber capacity has to be extracted in 15 to 20 seconds in order to simulate a typical model climb. On completion of the climb simulation, a slight increase in pressure is required in order to emulate the transition to gliding flight.

For verification purposes it is only necessary to perform tests starting at local ambient pressure. Slight air leakage of the chamber is acceptable and aids the gliding flight simulation.

Within the chamber, facilities are required to provide power and simultaneous command signals to a number of AMRTs.

A known, good AMRT (preferably with a certificated calibration) is used as the reference standard against which other AMRTs are compared.

The procedure is as follows.

- (a) Place the device(s) to be tested in the chamber together with the reference device.
- (b) Readout display(s) may be attached either before or after the pressure change has been applied.
- (c) Close the chamber and allow a short period for the pressure to stabilise. (Some variation may occur due to temperature differences introduced by handling).
- (d) With the Motor Command set to “motor off”, apply power to the AMRTs. This causes the zero height reference to be detected by the AMRT(s).
- (e) Apply a Motor Run command.
- (f) Reduce the chamber pressure at an appropriate rate to simulate a climb to approximately 150 metres.
- (g) Apply a Motor Stop command and immediately allow the pressure to begin increasing to simulate a descent of gliding flight.
- (h) Wait for a minimum of 10 seconds.
- (i) Observe ‘Start Height’ readings for all AMRTs.
- (j) Analyse the results by comparison with the known, good AMRT.

Any AMRT that deviates by more than plus or minus 1.25 metre from the monitoring AMRT should be subject to further investigation.

Note: The overall tolerance for absolute accuracy specified in Section 2 is the worst case figure when all possible discrepancies are taken into consideration.

(The 1.25 metre figure may be subject to revision when more experience has been gained in the competition environment).

Repeat the procedure simulating other climb heights as considered necessary.

6.0 GUIDANCE FOR COMPETITION ORGANISERS ON THE USE OF MONITORING AMRTS

Competition rules allow competition organizers to carry out random checks on competitors' equipment at any time during a competition. One aspect of this could be the use of a monitoring AMRT to be fitted in a competitor's aircraft for a competition flight. Item 4.1.9 defines that the appropriate connectors in an aircraft must be accessible.

6.1 Use of Monitoring AMRT in Competitors' Aircraft

- 6.1.1 A monitoring AMRT may be fitted at the command signal input of an AMRT by means of a Y connection lead. This permits the monitoring AMRT to respond to the motor command signals and take its own 'Start Height' measurement of a flight.

The rating and quality of the Y lead must be such that it does not degrade the performance of the competitor's own installation.

- 6.1.2 The 'Start Height', as determined by the monitoring AMRT should be in close agreement with the reading taken by the competitor's own AMRT.

Allowing for variations in pressure distribution within the aircraft, agreement within plus or minus 2.0 metre is considered acceptable.

Note: The 2.0 metre figure may be subject to revision when more experience has been gained in the competition environment.

7.0 APPROVAL PROCEDURES

7.1 Check list of features to be verified when evaluating compliance

- (a) Display of firmware revision level.
- (b) Operational sequence (as in 5.2.2).
- (c) Retention of Start Height reading when power removed.
- (d) Accuracy of Motor On pulse sensing.
- (e) Accuracy of timing of motor run.
- (f) Accuracy of 10 second period after motor cut.
- (g) Accuracy of ISA calculation across full ambient pressure range.
- (h) Accuracy of Start Height determination across range of Start Heights.
- (i) Dynamic response of Start Height determination.
- (j) Effect of temperature variations on Start Height accuracy.
- (k) Prevention of motor restart.
- (l) Clarity and legibility of readout.
- (m) Verification that any additional features not required by the specification (eg buttons on the display, flight logging) do not affect the essential operation of the device.

7.2 Submission of AMRTs for Approval

The CIAM is establishing a Working Group to assume responsibility for the approval process. The services of a technical expert may be used to undertake the practical testing on behalf of the Working Group.

Further details will be advised in due course, together with the specific requirements concerning the presentation of devices for approval.

Devices submitted for approval must be to normal production standard and must incorporate firmware at the revision level that is to be evaluated.

Supporting documentation shall include the sensor manufacturer's data sheet on the pressure sensor device with particular reference to long term drift and temperature variation characteristics.

Additional supporting information may be required at the request of the technical expert undertaking the evaluation on behalf of the CIAM Working Group.

Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.

7.3 Withdrawal of Approved Status

- 7.3.1 An approved AMRT may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the AMRT.
- 7.3.2 If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the AMRT, the validity of all AMRTs on the Approved List will be subject to review.

Section 2 - Technical Guidance Notes and Technical Specification for Altimeters Used in Space Modelling Competition V 2.0

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4.0 APPROVAL OF ALTIMETERS

1.0 INTRODUCTION

1.1 Purpose of this Document

The FAI CIAM rule making/amendment process is geared to maintain competition rule stability and as such presents some difficulties in areas subjected to rapidly changing technologies. To overcome this problem, the CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” working group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of altimeters for use in Space Model contests, where places in several events are determined based on the maximum flight altitude achieved by models. The information contained is relevant to altimeter manufacturers, competition organizers, competitors and persons involved in the evaluation of Space Model altimeters.

1.2 Philosophy behind the Altimeter Specification

Certain classes within the FAI structure of Space Model contests require the measurement of flight altitude. Such events attract large numbers of competitors each of whom is entitled to multiple flights. Consequently, competition procedures require organizers to deal with a significant number of altimeters for operation and verification purposes.

- a) To accurately determine the ‘maximum altitude achieved’ as required for the competition.
- b) To ensure that the altimeter can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- c) To provide a standard method of operation for use of altimeters in a competition.

2.0 MANUFACTURING SPECIFICATIONS

This section is the formal specification for altimeters intended for use in the space modelling competition.

2.1 Altimeter Technical Specification

The essential requirements of an electronic altimeter carried in a space model are that it must:

- a) Use a barometric pressure measurement technique.
- b) For a single flight, measure and record the difference between the altitude achieved at each moment of sampling after launch and the altitude of the pad from which it was launched.
- c) Retain the flight data, even with power removed.
- d) Be equipped with a data communication interface to be used for accessing data and managing the altimeter as described in Section 2.1.3.2.
- e) Provide an audible and/or visual indication of its operating status. For the purposes of this document 'flashes' shall mean audible beeps and/or visual flashes.
- f) Have no configurable settings or adjustments that allow the user to modify its operation such that it will operate outside the requirements of the competition rules.
- g) Be powered by a flight battery, either rechargeable or non-rechargeable, having sufficient capacity to operate the altimeter for a minimum of two hours before requiring recharging or replacement.
- h) Have dimensions that permit the altimeter, together with its flight battery, to fit in a cylindrical section of a space model airframe that is no greater than 17.5 millimeters in diameter and 30 millimeters in length.
- i) Mass not exceed 3.5 grams in weight, with flight battery.

2.1.1 Barometric Pressure Measurement

- a) The conversion of pressure measurements to height information must be based on the International Standard Atmosphere, as defined in ICAO Document 7488/2. The detailed requirements are specified in Section 2.2 of this document.
- b) The method of converting pressure measurements to height information performed by the altimeter firmware must be such that the specified accuracy is maintained up to a height of 4000 meters above sea level.
- c) Within the altimeter system, it must have a smoothing or filtering technique that enables it to recognize and reject any abrupt pressure transients that are the result of flight dynamic events such as staging, activation of a rocket motor ejection charge, or wind gusts.
- d) It must have a measurement accuracy of better than 2 meters.
 - 1) This accuracy must be achieved across an ambient ground level pressure range of 750 to 1050 hPa and across an ambient ground level temperature range of 0 to +50 degrees Celsius.
 - 2) This accuracy shall include the effects of the manufacturer-specified accuracy of the pressure sensor. Such effects include:
 - The resolution of the analogue to digital conversion
 - Signal noise
 - Thermal drift across a period of two hours of operation when operated at 15 degrees Celsius
 - Hysteresis effects over the specified working range of the altimeter
- e) The calculation of pressure shall incorporate any calibration parameters provided by the pressure sensor manufacturer.
- f) Temperature measurements made by the pressure sensor device must not be used in the calculation of pressure to altitude, except in accordance with the sensor manufacturer's recommendations for sensor device compensation purposes.

2.1.2 Operating Parameters

- a) Height data shall be recorded and indicated in meters with a precision of one decimal place.
- b) The working range shall be no less than 2000 meters when operated at launch site altitudes up to 2000 meters above sea level.
- c) The data sampling rate shall be not less than ten samples per second.

2.1.3 Sequence of Operation

- a) The altimeter manufacturer shall provide a technical report describing the sequence of operation for the altimeter. The technical report shall include a table describing the indication method (blink sequence, etc.) for the operating modes. For continental and world championships, the altimeter technical report shall be provided to all competitors at least 60 days before the competition.
- b) Following the application of power, the altimeter shall determine whether it is in a Flight Recording Mode or connected to a readout device in Data Communication Mode.

2.1.3.1 Flight Recording Mode

- a) On application of power, the status of the altimeter shall indicate in a manner that can be determined by the competitor that
 - 1) It is correctly reset and ready to accept new flight data; or 2) it contains flight data and has not been reset. An altimeter that has not been reset must not be used for flight until a reset has been applied.
- b) On application of power, a correctly reset altimeter shall:
 - 1) Enter a preparation period for the purpose of installing the altimeter in a space model. During the preparation period, the altimeter shall not record data so that the altimeter's results will not be affected by handling or other preparation activities.
 - 2) The preparation period will conclude after a specified time or by other procedure as defined in the altimeter technical document.
- c) On completion of preparation period, the altimeter shall become ready for collection of flight data and become ready to detect a launch. If flight readiness is activated by an external command, the success of this external command shall be communicated to the RSO prior to launch.
Once the altimeter is in a flight-ready state the sampling of air pressure shall commence at the sampling rate defined in 2.1.2.c).
- d) Lift-off and commencement of a flight is considered as having been detected when a flight altitude of at least 30 meters has been achieved.
- e) When a flight has commenced, the altimeter shall:
 - 1) Use the calculated flight altitude data to determine the peak altitude above the launch pad achieved during the flight. This value must be recorded and retained for subsequent retrieval.
 - 2) Retain recorded Flight Data for the period commencing at least 5 seconds before lift-off, until at least 30 seconds after lift-off.
 - 3) Change the status indication to indicate flight data has been recorded.
- f) The altimeter shall remain in this state until power is removed.

2.1.3.2 Data Communication Mode

Data Communication Mode is required for retrieval of flight data and resetting of the altimeter.

The use of Data Communication Mode is the only permitted method of resetting the altimeter.

In Data Communication Mode the audible/visual indicator will indicate in accordance with the altimeter technical document.

The altimeter must connect, physically or by radio-frequency signal, with a digital device such as a computer equipped with a software package that will:

- a) Display the firmware revision level of the processing firmware loaded onto the altimeter.
- b) Display a unique serial number allocated to the altimeter by the manufacturer.
- c) Display the peak altitude above the launch pad achieved during the flight, as determined by the altimeter.
- d) Display the full digital output of the altimeter's altitude versus time data in graphical or tabular format.
- e) Permit resetting the altimeter so that:
 - 1) Any previously recorded flight data is either deleted or moved to a file that is clearly recognizable as a flight made prior to reset.
 - 2) The altimeter becomes initialized so that when subsequently powered up in Flight Recording Mode, it will initiate a new flight data collection sequence.
 - 3) Following application of the reset, the audible/visual status indication of the altimeter shall be as defined in the altimeter technical document.

2.1.3.3 Additional Features

The hardware revision level must be printed on the PCB surface

2.2 Altitude Calculation

Pressure at a given location is caused by the weight of the atmosphere above. This varies with air density, humidity, gravity and temperature. In addition, temperature itself reduces as height increases. Consequently, some form of standardized definition is required to enable instrumentation based on barometric pressure to be used to determine height accurately.

The International Standard Atmosphere (ISA) assumes a mean sea level pressure of 1013.25 hPa at a temperature of 15 degrees Celsius, a specified temperature lapse rate with height, a specified gravitational force and a given molecular density of air.

Conditions on any day are unlikely to correspond exactly with the ISA model; a typical spread of 50hPa can occur at a given location due to normal meteorological conditions.

Because records for altitude may be set in space modelling competitions, it is necessary to apply an adjustment for launch site ambient temperature to the standard ISA calculated value in order to achieve greater absolute accuracy of the calculated altitude.

- 2.2.1 In the lower atmosphere (up to 12km), the 15 degree Celsius ISA relationship for a pressure P, can be described by the following formula:

$$\text{ISA Altitude} = K1 \times \{1 - (P/P_0)^{K2}\} \text{ meters where}$$

$$K1 = 44330.76923 \text{ meters}$$

$$K2 = 0.190266669$$

$$P_0 = 1013.25 \text{ hPa (15 degree Celsius ISA sea level pressure)}$$

K1 is derived from the formula:

$$K1 = T/L$$

Where T = temperature in degrees Kelvin

(15 degrees C is equivalent to 288.15 degrees Kelvin)

L = Lapse Rate 0.0065 degrees K per meter.

K2 is derived from the universal gas constant, temperature lapse rate, gravitational force, molecular mass of dry air and temperature with respect to absolute zero.

This formula results in a relationship between pressure and height that is represented by a curve that is steeper at high pressure/low altitude and less steep at low pressure/high altitude.

Note: In this context, results from calculations using the formula of the 1976 US Standard Atmosphere, agree closely with those of the ISA formula in the lower atmosphere and are considered acceptable over the space modelling working range.

- 2.2.2 The calculation of the altitude achieved by a space model requires a precise measurement of the difference between two ISA-derived pressure altitudes. The first is the zero reference value at launching pad level. The second is the altitude that is measured during each moment of the model's flight.
- 2.2.3 In order to achieve sufficient accuracy for space modelling competition, the method of firmware processing of pressure data is important. Whilst ideally the ISA computation should be performed by an arithmetic calculation with maximum precision, processing constraints may make this impractical. It is permitted to use an algorithm approach to emulate the computation provided it can be demonstrated that an appropriate accuracy is maintained and no discontinuities are introduced in the ISA curve.
- 2.2.4 Altimeter manufacturers may wish to adopt pressure sensor modules that have an inbuilt pressure-to-height conversion. In such a case, it must be demonstrated that the computation complies with the accuracy requirements over the specified working range of the space modelling application.
- 2.2.5 In order to compensate for the effect of launch site ambient temperature on the conversion of pressure to ISA altitude, a factor K3 must be applied to the 15 degree Celsius ISA value. This correction is to be applied post-flight using the temperature recorded by the competition organizer for the round in which the flight occurred.

$$\text{Temperature Corrected Altitude} = K3 \times \text{ISA Altitude}$$

$$\text{where } K3 = (273.15 + T_a) / 288.15$$

Ta = ambient temperature in degrees Celsius

3.0 COMPETITION PROCEDURES

This section provides procedures for the competitor and for the competition organizer for employment of altimeters in competition.

3.1 Competitor Installation of Altimeters

- 3.1.1 The installation of the altimeter in the space model is the sole responsibility of the competitor, who must ensure that the altimeter is not subjected to conditions that might cause it to fail or to operate in a manner different from that required for the rules of the competition.
- 3.1.2 Pressure sensor devices can be affected by strong light or heat falling on the sensitive area. Therefore the altimeter must be installed in the space model in such a manner that the sensitive area is shielded from strong light levels or heat.
- 3.1.3 The altimeter must be placed inside the space model in the state as supplied by the contest officials. The use of any method that modifies the true barometric pressure at any time is prohibited.
- 3.1.4 The altimeter shall be shielded from direct exposure to the hot and corrosive gases of any rocket motor or its ejection charge.
- 3.1.5 The compartment in the space model that houses the altimeter must fully enclose it through apogee of the flight; must permit its removal after flight but not permit its separation from the space model during flight; and must be provided with a vent hole or holes of a size sufficient to rapidly equalize its internal pressure with the ambient pressure at the altitude of the model throughout its flight.
 - a) If the altimeter communicates externally via radio-frequency signal, the compartment should have no metallized surfaces or carbon construction.
 - b) The vent holes must be located on a straight surface, not the curved portion of a model's surface at the nose. The vent holes must be burr-free and not located behind any protrusions from the surface.

3.2 Organizer Management of Altimeters

- 3.2.1 All altimeters used in an altitude competition shall be of the same type and shall be provided by the competition organizer.
 - a) It is recommended that the competition organizer tender offers from more than one producer whose devices are approved as fully meeting the technical standards of this code and announce to all competitors the type that has been selected for use at least 60 days in advance of the competition.
 - b) The competition organizer shall provide a number of altimeters at least equal to 1.2 times the number of registered competitors in the altitude classes for that contest.
 - c) The producer shall provide written instructions in English along with appropriate digital device connectors and software for the organizer's pre-flight set up and post-flight readout of the altimeters and for the competitors' use of the altimeters.
 - d) The producer shall calibrate all altimeters at his premises and shall provide the organizer with calibration certificates for each altimeter, by serial number
 - e) The producer shall, if required, provide an altimeter test apparatus for the organizer, and the organizer shall use such an apparatus to conduct a verification that the accuracy of the altimeters is in compliance with 2.1.1 before the beginning of the competition.
 - 1) The apparatus shall allow verification of all altimeters simultaneously, and shall be used for all altimeters at three test altitudes: 300 meters, 600 meters, and 1000 meters.
- 3.2.2 The competition organizer shall collect a deposit for each altimeter that is issued during the competition, sufficient to cover the cost of the altimeter and its battery; shall return this deposit if the altimeter is returned in working order; shall keep the deposit for altimeters not returned or returned with damage that prevents their continued use.
- 3.2.3 The competition organizer shall maintain a log and record the serial number of each altimeter issued to a competitor during the competition and the start number of that competitor; shall record the time of issue and of return of each altimeter; and shall collect initials from both the organizer's representative and the competitor (or the competitors team manager) for each issue and each return of that altimeter during the competition.

3.3 Employment of Altimeters in the Competition

- 3.3.1 All altimeters that will be used in a competition shall be kept in the secure custody of the competition organizer in a locked box out of direct sunlight once they have been accepted following verification of calibration, except when issued to competitors or team managers during the competition.
- 3.3.2 Altimeters shall be issued during the competition from a single Altimeter Control Station with two Altimeter Controllers: one to set and issue the altimeters and one to maintain the log required in 3.2.3 and collect the deposit required in 3.2.2.
- 3.3.3 Altimeters shall be issued to and used by competitors once a flying round has begun in accordance with the following procedure:
- a) The Altimeter Controller will observe and record the ambient temperature of the launch area at the beginning of each flight round and will write that observed value on the flight cards for all altimeters that are issued during that round.
 - b) The competitor will insert the rocket motor(s) in his space model and otherwise prepare it for flight.
 - c) The competitor will proceed to the Altimeter Control Station with his model and flight card, in company with a launch site monitor who has a stopwatch.
 - d) The Altimeter Controllers will issue an altimeter selected at random from a common pool, with a charged battery, after verifying that the altimeter has been reset for a new flight, and will write the serial number of this altimeter on the flight card.
 - e) The launch site monitor will start his stopwatch when the Altimeter Controller connects the altimeter to its battery and will observe the competitor insert the altimeter into his model within the preparation period before the altimeter goes into its data collection mode, and will then accompany the competitor back to the launch lane.
 - f) The competitor will fly and recover the model and will present it to the launch site monitor upon recovery.
 - g) The competitor will return to the Altimeter Control Station with the model and flight card in company with the launch site monitor.
 - h) The Altimeter Controller will read and save the altimeter's full recorded flight data including serial number on a computer, apply the ambient temperature correction required in 2.2.5, and write the resulting score on the flight card after verifying that the altimeter serial number matches the serial number issued before the flight, present a score report showing the altitude recorded by the altimeter to the competitor for his signature and provide a second copy of that report to the competitor, then reset the altimeter following that signature and return it to the common pool for future issue.
 - i) If the competitor disputes the score and declines to sign the score report the altimeter will be kept separately from all others and its data will not be reset until directed by the FAI Jury.
- 3.3.4 If the altimeter experiences an anomaly, in use that puts the accuracy of its data in doubt or causes it to provide no data, then the flight shall be scored at a "Track Lost" (TL) and the competitor may re-fly if there is enough time to perform another flight in the same round.
- a) Such anomalies include evidence that the altimeter may have been tampered with or removed during the recovery process prior to presentation to the launch site monitor, or failure of the altimeter to initiate a data-collection sequence during flight.
 - b) Loss or damage of the altimeter during flight shall be a TL but shall not be the basis for another flight attempt in the same round.
 - c) Failure to return the model with altimeter to the launch site monitor within 30 minutes after the end of the final round of the event shall be a TL.

4.0 APPROVAL OF ALTIMETERS

- 4.1 The CIAM has established a committee to assume responsibility for the approval process for altimeters that are designed in accordance with the requirements of Section 2 of this document and may be eligible to be used in Space Modelling competition. The services of a technical expert may be used to undertake the practical testing on behalf of the committee.
- 4.2 A manufacturer seeking Devices submitted for approval must be manufactured to normal production standards and must incorporate altimeter firmware and supporting digital device software at the revision level that is to be evaluated.
- 4.3 The altimeter manufacturer shall submit the evaluation form “SM-ALT_Compliance_Matrix_Template.xlsx”. Any non-compliance items or request for waivers shall be highlighted.
- 4.4 The altimeter manufacturer shall submit a technical document describing the altimeter system including hardware features (sensors, memory, etc.) and software methodology (sampling rate, filter methods, etc.). Supporting documentation shall include the sensor manufacturer’s data sheet of the pressure sensor device with particular reference to long term drift, hysteresis, and temperature variation characteristics.
- 4.5 Additional supporting information may be required from the altimeter manufacturer at the request of the technical expert undertaking the evaluation on behalf of the CIAM committee.
- 4.6 A list of approved altimeters, together with copies of approval documents, is published on the CIAM website under the section ‘EDIC WG & Electronic Device Approvals’. Updates are made immediately any new approval is issued. Competitors and Contest Organizers are advised to regularly check for updates to the list.
- 4.7 Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.
- 4.8 An approved altimeter may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the altimeter.
- 4.9 If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the altimeter, the validity of all altimeters on the Approved List will be subject to review.

Section 3 – F3E Energy Limiter V 1.1

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1.0 Introduction

1.1 Purpose of this Document

The FAI CIAM rule making/amendment process is geared to maintain competition rule stability and as such presents some difficulties in areas subjected to rapidly changing technologies. To overcome this problem, CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” Working Group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of F3E Energy Limiters (F3EELs) for use in F3E competitions. The information contained herein is relevant to F3EELs manufacturers, competition organizers, competitors and persons involved in the evaluation of F3EELs.

1.2 Philosophy behind the F3EEL Specification

International competitions attract large numbers of competitors, each of whom is entitled to three models. Consequently, competition organizers could expect to be presented with a significant number of F3EELs from a number of different manufacturers for scrutinizing and verification purposes.

It is against this background that the F3EELs Specification has been formulated with the following objectives:

- a) To accurately determine the consumed energy as required for the competition.
- b) To ensure that the F3EEL can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- c) To provide a standard method of operation so that, irrespective of F3EEL manufacture, verification testing can be performed easily with a defined test procedure.
- d) To define standard connectors for compatibility with test facilities and to reduce the risk of accidental damage during handling by competition officials.
- e) To provide a standard method of determining that the F3EEL is of the type for which an approval has been granted.
- f) To guarantee the particular version for which the device was approved.

2.0 INFORMATION FOR MANUFACTURERS

2.1 F3EEL Technical Specification

This section is the formal specification for F3EELs intended for use in the F3E competition environment.

2.2 Technical Specifications

An electronic F3EEL carried in an F3E model shall fulfill the following technical specifications:

- a) Current measurement will be by use of resistive shunt resistor or current sensing Hall sensor type device installed in the positive side of the power lead between the battery and the motor controller.
- b) Voltage measurement will be made between the positive and negative leads of the battery that supplies power for the motor. Alternative to negative battery lead is the negative lead of the ESC Rx connector.
- c) The method of converting current and voltage to energy units (Wmin) must be such that the specified accuracy is ensured over the operating range. The calculation will be Watts (Volts x Amps) x minutes motor run.
- d) The motor will be stopped once the accumulated total energy limit of 1000 Wmin has been reached by output of a signal to the ESC of less than 1.2 ms. The motor cutoff will continue for 10 seconds after which additional motor run time will be allowed until the additional 10% energy limit is reached at which point another 10 second motor cutoff will occur. This sequence shall repeat until the model lands. Alternatively, the motor will be stopped once the accumulated total energy limit of 1000 Wmin has been reached by output of a signal to the ESC of less than 1.2 ms for 10 seconds after which the motor can be controlled normally if the Rx signal has been set to 1.0 ms for a moment.
- e) The F3EEL must have compatibility with ESCs using the nominal 1 to 2 milliseconds (ms) control signal from the receiver such that no low speed motor drive occurs when “motor stop” is instituted either by the competitor or the F3EEL.
- f) The supply of power to the F3EEL shall be that available from the receiver output to the ESC at the receiver motor command connector. If the power source to the receiver is supplied via the ESC motor command connector, it is permissible for this to be supplied to the receiver via the F3EEL.
- g) The connectors of the F3EEL shall be of a universal type compatible with JR/Futaba and must be proof against any misalignment that could result in electrical damage to the F3EEL or connected equipment. Where an extension lead is used to satisfy this requirement it must be secured to the F3EEL in such a manner that it cannot be unintentionally removed.

2.3 Characteristics of the F3EEL

- 2.3.1 The accuracy of the determined energy (Wmin), shall be within the range +/- 2%. This accuracy must be achieved across input voltage between 13 and 21 volts and across 25 to 130 amps and ambient temperature range of +10 to +50 degrees centigrade.
- 2.3.2 The calculation of the energy shall incorporate any calibration parameters provided by the current sensor manufacturer.
- 2.3.3 The sampling rate of the power measurement used for energy usage determination shall be a minimum of 10 samples per second.
- 2.3.4 There must be no configurable settings or adjustments that allow the user to modify the operation

of the F3EEL such that it will operate outside the requirements of the competition rules.

- 2.3.5 The Rx output signal of the F3EEL must follow the Rx input signal as accurate as $1\mu\text{s}$ unless the stop condition is met.
- 2.3.6 Accumulating energy (W_{min}) must continue when motor stop condition is asserted unless the current drawn by the motor is less than 5 Amps. This is to avoid a delayed motor off by the ESC.

2.4 Starting condition

The F3EEL shall initialize using the following procedure:

- (a) Immediately on application of power to the F3EEL it will zero out the energy accumulator.
- (b) The motor command will initialize to the “off state” and will remain there until the radio control command commands otherwise.
- (c) When the motor start command exceeds 1.2 ms the F3EEL will start to accumulate W_{min} .
- (d) When the measured current exceeds 5A the F3EEL will also start to accumulate W_{min} .

Notes:

- 1) A Motor Start command is defined as any control signal exceeding 1.2 milliseconds.*
- 2) When the limit is reached refer to 2.2 d) for information on operation.*

3.0 TECHNICAL CONSIDERATIONS

This section is provided as design guidance on F3EELs intended for use in the F3E competition environment.

3.1 Restriction on User Configuration Options

Whilst for commercial considerations manufacturers may wish to produce F3EELs adaptable for differing applications, when used in the F3E competitive environment, the installed firmware shall have dedicated F3E functionality with no user configurable options. The purpose of this is to eliminate any possibility of errors in operation due to incorrect configuration settings.

3.2 Connector Compatibility

The F3EEL shall use standard JR/Futaba type connectors for both the signal in and outputs.

The power connectors will be 6mm connectors with a female on the positive lead from the battery and a male on the negative lead from the battery. The F3EEL will be fitted 6mm connectors so as to plug in between the positive ESC lead and the positive battery lead. Negative ESC plug will be plugged into the negative battery plug. The ground reference for the F3EEL is through the ESC ground shared with the RC signal (see 2.2b).

3.3 ESC Compatibility

There is no defined standard for the manner in which ESCs (electronic speed controllers) initialize to the range of motor command signals. In addition, programmable transmitters allow the user to set the motor command signals to any value within the programmable range.

- 3.3.1 The F3EEL specification of motor on threshold at 1.2 milliseconds (ms) has been selected as being compatible with most known equipment but cannot be a guarantee of correct operation for every combination of transmitter, receiver and ESC.

4.0 COMPETITION INFORMATION

4.1 Installation Environment

- 4.1.1 The installation and functioning of the F3EEL, the transmitter and its programming, receiver and ESC, is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.
- 4.1.2 The F3EEL must be installed in such a way that it is protected from mechanical damage.
- 4.1.3 The F3EEL must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the current or voltage sensing at any time is prohibited.
- 4.1.4 The ESC must always operate via its series connection to the F3EEL and not with a direct connection to the receiver.

5.0 VERIFICATION OF F3EELS

This section details procedures for verifying the operation of Approved F3EELS for use in the F3E competition environment. It is provided as a guide for competition organizers.

5.1 Purpose of Verification

All F3EELS used in an F3E competition are required to be of an Approved Type. The verification process serves to ensure that no individual F3EEL operates in a manner different from others in use at the competition.

5.2 Verification Testing of 1000 Wmin F3EEL

5.2.1 Equipment required for testing.

- 1) A regulated power supply capable of 0 to 25 volts output.
- 2) A current source capable of up to 100 amps.
- 3) A constant current electronic load capable of up to 100 amps.
- 4) Suitable meters and shunts for reading current and voltage with 1% or better accuracy.
- 5) Stopwatch capable of at least 0.1 second resolution.
- 6) A “servo simulator” to generate a motor command pulse.
- 7) Suitable meter or oscilloscope to monitor motor command output of F3EEL.

5.2.2 Verification testing will be carried out at a nominal 1000 Watts. The voltage for testing will be 20 volts and the current will be 50 amps.

5.3 Test sequence for checking Wmin accuracy.

- 1) A 1000 Watt simulated load will be applied to the F3EEL using the above test equipment.
- 2) The stopwatch will be started when the motor command pulse is changed to above 1.5 ms.
- 3) The motor control pulse out of the F3EEL will be monitored and when it changes to below 1.2 ms the stopwatch will be stopped and the 1000 Watt load terminated. The time will be recorded.

5.4 Firmware Revision Level

All F3EELS will have a label on the exterior surface. The label information will include the name of the manufacturer, model number and firmware version.

5.5 Test Pass/Fail limits

- 5.5.1 An F3EEL is considered to have passed if it reduces the motor command to below 1.2 ms when the 1000 Watt load has been applied for 1 minute +/-2% (58.8 to 61.2 seconds).
- 5.5.2 An F3EEL is considered to have passed if all the test requirements outlined in this document are satisfied.

6.0 Submission of F3EEL Devices for Approval

The CIAM is establishing a Working Group to assume responsibility for the approval process. The services of a technical expert may be used to undertake the practical testing on behalf of the Working Group. Further details will be advised in due course, together with the specific requirements concerning the presentation of devices for approval.

Devices submitted for approval must be to normal production standard and must incorporate firmware at the revision level that is to be evaluated.

Supporting documentation shall include the sensor manufacturer's data sheet on the pressure sensor device with particular reference to long term drift and temperature variation characteristics. Additional supporting information may be required at the request of the technical expert undertaking the evaluation on behalf of the CIAM Working Group.

Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.

6.1 Withdrawal of Approved Status

An approved F3EEL may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the F3EEL.

If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the F3EEL, the validity of all F3EEL devices on the Approved List will be subject to review.

Section 4A – Performance Requirements Definition for F5B Real Time Energy Consumption Telemetry (REAL-TECT) System Compatible with Existing “GASSENSOR” Equipment

Initial Publication: 01 July, 2016

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- 6.0 Submission of REAL-TECT for Approval
 - 6.1 Withdrawal of Approved Status

1.0 Purpose of this Document

The FAI CIAM rule making/amendment process is devised to maintain competition rule stability and as such presents some difficulties in areas subjected to rapidly changing technologies. To overcome this problem, CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” Working Group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of Real Time Energy Consumption Telemetry Systems (REAL- TECT) for use in F5B competitions. The information contained herein is applicable to F5B electronics equipment manufacturers, competition organisers, competitors and persons involved in the evaluation of REAL-TECT systems.

1.1 Previous Equipment used in F5B Competitions

Before the introduction of the telemetry equipment described in this document, the electronic determination of whether or not a contestant’s motor is running, is based exclusively on signals coming from an extra ground based R/C receiver connected to a monitor known as a “GASSENSOR”.

This secondary receiver’s throttle channel is connected to the GASSENSOR and the GASSENSOR infers the motor is running when the pulse width of that channel rises to a threshold 100 μ S above that recorded when the throttle is low. The converse is also true as the motor is expected to be off when the throttle pulse width decays to below the threshold.

The GASSENSOR system was designed before the introduction of computer/microprocessor based Radio Transmitters and software programmable speed controllers. With these modern radio control systems and speed controllers there is no guarantee that the output of the secondary receiver truly reflects the operation of the motor in contestants’ models.

During the F5B Distance Task, the GASSENSOR communicates the detected Motor- ON/Motor-OFF events to the base station via wired RS-232. At the conclusion of the Distance task, the GASSENSOR automatically commences to measure duration time, continuing the timing operation until duration ends.

The GASSENSOR is disconnected from the base station (and transported by an official) in order to stop duration time at the landing circle.

The length of the Motor runs during duration is also accumulated using this inferred Motor-ON logic.

1.2 Philosophy of the F5B REAL-TECT System

The REAL-TECT system shall use motor current obtained as the primary mechanism to infer the motor is running and provide that information to the GASSENSOR via telemetry.

The REAL-TECT ground station shall connect between the extra receiver and the GASSENSOR, maintaining all functionality of the GASSENSOR.

1.3 F5B REAL-TECT System Functionality

The REAL-TECT system shall directly read the motor current as a measure of whether the motor is running and transmit that information via RF to the Ground Station and Contest Organisers in real time. There shall also be a mechanism for a second R/C receiver bound to a contestant’s transmitter to be used only as backup in case of RF system failure or RF out of range condition.

The REAL-TECT ground station shall constantly evaluate the condition of the RC based motor signal and the RF based motor signal in order provide the most conservative input to the GASSENSOR; any ambiguities will be construed as motor not running.

There shall be two connections to the contestant’s airborne equipment: one in series with battery positive lead to supply power to the airborne device and to monitor motor current; the second to the battery

negative connection to supply ground.

The GASSENSOR will receive the adjudicated Motor Status signal via the REAL-TECT ground station as if it were an R/C throttle pulse high or low so that any Course motoring foul may be applied in real time. The pilot may then adjust his flight strategy in accordance with the Class rules.

Near real time WM telemetry information replacing an absolute 1750 WM Limiter will allow pilots to better apportion their energy use between Distance and Duration and to know that a penalty is being incurred once 1750 WM is attained. A known and calculable penalty after 1750 WM should be preferable to the possibility of a lost model due to unforeseen cessation of power while being in a precarious or unsafe flight condition and having no ability to apply power to correct the condition and/or return to the field.

International competitions attract large numbers of competitors, each of whom is entitled to two models thus a significant amount of REAL-TECT qualified equipment is necessary and must be seamlessly integrated with organizer owned "GASSENSOR" equipment.

Consequently competition organizers may be offered REAL-TECT systems from a number of different manufacturers for qualification, verification and/or revenue purposes. It is against this background that the REAL-TECT Specification has been formulated with the following objectives:

- a) To accurately measure, tally and broadcast via RF, the consumed energy used by the model as required for the competition.
- b) To accurately detect presence of current flowing from the model's motor battery to the speed control and broadcast the event via RF.
- c) To accurately decode the Motor-On RF event and adjudicate it with the Motor-On RC event and provide the net result to the GASSENSOR in real time as equivalent to only an R/C Throttle Pulse High signal received from a parallel R/C receiver.
- d) To accurately receive and decode the RF WM tally and a mechanism to convey it to the pilot's helper on a demand basis.
- e) To ensure that the REAL-TECT system can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- f) To provide a sufficiently standard method of operation so that, irrespective of REAL-TECT manufacturer, verification testing can be performed easily with a defined test procedure.
- g) To define standard connectors for compatibility with test facilities and to reduce the risk of accidental damage during handling by competition officials.
- h) To provide a standard method of determining that the offered REAL-TECT system is of the type for which an approval has been granted.
- i) To guarantee the particular hardware and software version for which the devices are approved.

2.0 Information for Manufacturers

2.1 F5B REAL-TECT Technical Specification

This section is the formal specification for REAL-TECT equipment intended for use in the F5B competition environment.

2.1.1 REAL-TECT System General Description

A manufacturer supplied REAL-TECT system shall consist of two major Components: Airborne Telemetry Transceiver (ATT) and Ground Display GASSENSOR Transceiver (GDGT).

A World Championship capable system will consist of as many GDGTs as GASSENORS as required for use in the competition and two ATTs per pilot. Sufficient spares will be required of each device.

There must be no configurable settings or adjustments that allow the user to modify the operation of the REAL-TECT system components such that a system can be operated outside the requirements of the competition rules.

The highest priority of the system is to detect motor current (in excess of a nuisance threshold) equating to Motor-ON from the telemetry system, adjudicate that with information from the second R/C receiver bound to a contestant's transmitter (referred to as "RC System") and communicate the motor status as quickly as possible to the GASSENSOR.

The lower priority system function, whilst the motor is stopped for greater than 1.5 seconds, is to convey to the pilot accumulated WM, alternately with Motor-OFF.

The transmissions taken in total shall constitute a "heart beat" message to the GDGT so it may constantly validate the RF condition and properly and seamlessly use it in conjunction with the Receiver R/C pulse value, to derive the proper GASSENSOR input.

Table 1 on the following page defines the required logic, 1 being Motor ON, 0 being Motor OFF and N being NULL or invalid signal:

RC-MOT-STATUS	RF-MOT-STATUS	GASSENSOR output pulse Width
N	N	1 ms <Motor OFF>
N	0	1 ms <Motor OFF>
0	N	1 ms <Motor OFF>
0	0	1 ms <Motor OFF>
1	0	1 ms <Motor OFF>
1	N	2 ms <Motor ON>
0	1	2 ms <Motor ON>
1	1	2 ms <Motor ON>
N	1	2 ms <Motor ON>

TABLE 1: R/C AND RF MOTOR STATUS LOGIC

2.1.2 REAL-TECT System RF Power and Frequency Band

The ATT shall have adequate RF power to have Line-of-Sight minimum nominal range of 500 metres. The GDGT is not required to transmit and if there is a facility for transmission, any adjustable RF power level should be set to minimum.

The RF frequency shall be within an ISM band allowed in each country, selected for adequate RF power. A typical ISM transmission allowed at +20 dBm in most Western European Countries is 433 to 438 MHz.

Use of 2.4 GHz equipment is strongly discouraged to prevent interference with the pilot's control of the model.

The RF modulation method, RF data rate and Communication Baud rate shall be at the discretion of the system designer subject to system performance requirements being met.

Each ATT shall be inalterably programmed with a unique serial number address, known as NODE_ID;

A block of serial numbers shall be reserved exclusively for contest use and a block for individual user experimentation.

REAL-TECT manufacturers offering equipment for competitions shall provide a list of NODE IDs offered for public sale/sold and those offered for unique use by competition organisers.

All ATT data transmissions shall include the unique NODE ID for positive source identification and a checksum for the GDGT to validate the data packet. In addition, if a fixed frequency, limited band, or software configurable frequency hopping mechanism is available, the unique NODE IDs should be spread over the available RF Channels.

An example of an eight Channel System using an 8 bit NODE ID is shown on the following page:

NODE ID	RF Channel
000-029	0
030-059	1
060-089	2
090-119	3
120-149	4
150-179	5
180-209	6
210-239	7
240-255	<Undefined>

TABLE 2: NODE ID and RF Channel

The GDGT must be configurable to accept RF signals from two ATT NODE IDs corresponding to a competitor's Primary model and to his Back Up model. These two NODE IDs must be on the same RF CHANNEL as detailed in Table 2.

The GDGT shall reject the Back Up NODE ID selection if it doesn't correspond to the same RF CHANNEL as the Primary NODE_ID.

Once either accepted ATT NODE ID transmits more than 50 WM, RF signals from the other NODE ID shall be ignored while the GDGT remains powered by the GASSENSOR.

2.1.3 Physical Arrangement: ATT

The ATT shall consist of a bi-directional current/voltage sensor and a processor- transmitter connected by a 40 ± 5 cm flat ribbon cable. The sensor may be either hall- effect, toroidal or shunt type and normally resides in the nose of the model convenient to the connection of the motor battery to the speed controller. The processor-transmitter resides in the fuselage immediately behind the wing.

The current/voltage sensor shall be connected in series with the positive battery connection and shall have a one 6 mm male barrel type connector and one female 6 mm barrel connector. The current/voltage sensor shall also provide system power ground via an additional 10 ± 1 cm lead terminated in 2 mm male barrel connector.

See Figure 1 on the following page for maximum allowed dimensions of the current sensor.

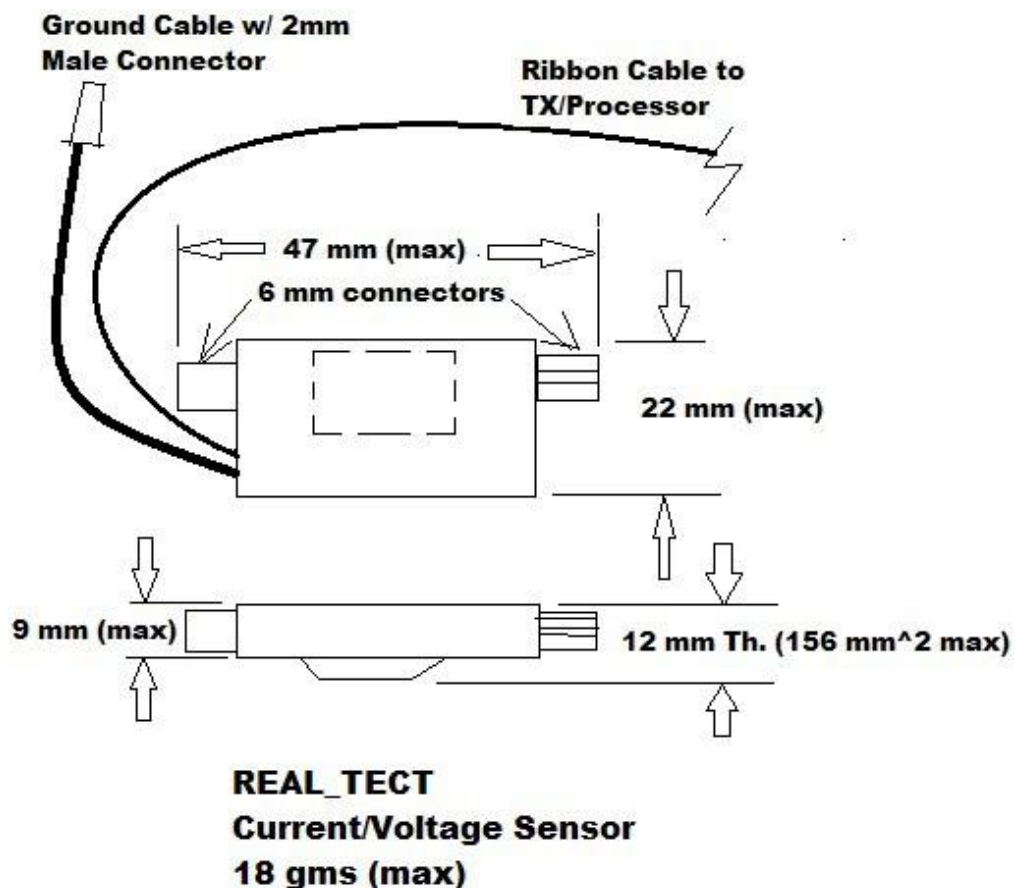


Figure 1: REAL-TECT CURRENT/VOLTAGE SENSOR

The processor-transmitter shall contain a microprocessor based circuit card assembly (CCA) that accumulates and stores the WM, an RF module having Country Approved frequency band/RF power and a suitable antenna.

The antenna may be integral to the module, directly connected to the module or connected to the module via micro-coax type cable of between 6 and 8 cm. It is permissible to specify that the cable exit the fuselage via a small hole or adjoining the wing opening in the fuselage.

The aerial configuration must be such that when the processor-transmitter is installed in a full carbon fibre model, the specified operating range requirements are satisfied.

See Figure 2 on the following page for maximum allowed dimensions of the processor- transmitter.

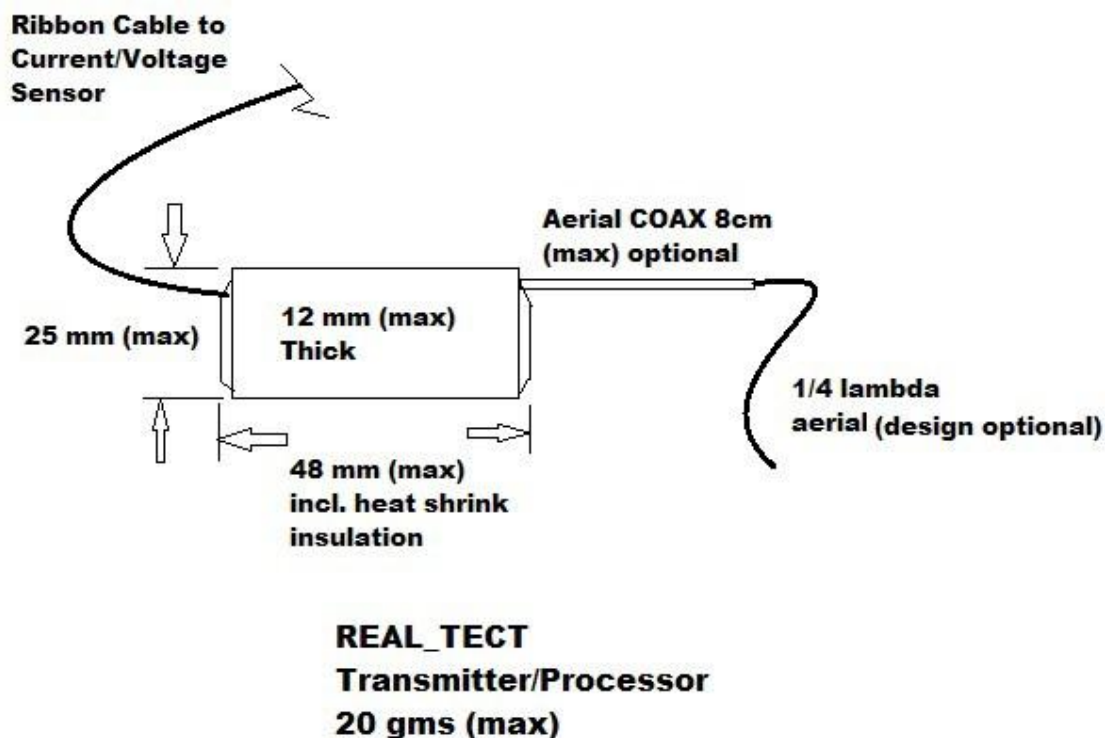


Figure 2 REAL-TECT TRANSMITTER/PROCESSOR

2.1.4 Physical Arrangement: GDGT

The GDGT shall have the following physical features:

- An enclosure containing all the equipment described below having maximum dimensions 15 cm X 10 cm X 5cm.
- A daylight readable alpha-numeric character display having a minimum of 32 characters.
- A set of buttons, momentary switches or other human input mechanism that configures the GDGT to the applicable NODE IDs.
- An ISM band Commercial-Off-The-Shelf (COTS) RF receiver configurable with any ATT and applicable Receiver Antenna.
- JR 3-pin female connectors to provide Motor Status to the GASSENSOR.
- JR 3-pin female connector to accept the motor signal from the Pilot's ground based receiver.
- The GDGT shall be powered by the GASSENSOR (4.6 V dc nominal) which will also power the contestant's receiver.

cont/...

2.1.5 Functional Description: GDGT Preparation for Flight

The GDGT shall be connected to the GASSENSOR via the GASSENSOR receiver port.

The pilot will provide his second Receiver to the operator who will connect the throttle pulse output to the GDGT Rx input port.

After powering on the GASSENSOR/GDGT/RC Receiver, the 32 character display (the DISPLAY) may show an opening message and must show its software version number for 2 +/- 1 seconds, prior to presenting a message requesting entry of the two NODE IDs corresponding to the ATTs of the Pilot's primary model and of the backup model. Without a backup model, the Primary NODE ID should also be stored as the Backup NODE ID.

After Node ID entry, the GDGT will continuously read and display the R/C throttle pulse value. The operator shall request the Pilot toggle his transmitter to RC throttle LOW if not already LOW. The GDGT will request operator make an input (e.g. press a button) to store the LOW pulse width.

The GDGT shall then show a message on the Display that the operator command the pilot to toggle his RC throttle to HIGH. The GDGT will request operator make an input (e.g. press a button) to store the HIGH pulse width. The operator shall ensure that the pulse value is stable before making the entry.

Once ENTER is pressed, the program will store the Backup NODE ID to volatile memory and if necessary change the RF channel of the RF Module.

While the Channel is being changed, or for 3 seconds, whichever is longer, the LCD will clear and then very briefly display "ADJUSTING RF CHANNEL TO <RF CHANNEL>".

The GDGT shall verify that

$\text{<Throttle High Pulse Width>} - \text{<Throttle Low Pulse Width>} = 600\mu\text{S}$ or greater and determine a value for motor on threshold ($\text{RC_MOT_ON_PULSE_WIDTH}$ as $\text{RC_MOT_ON_PULSE_WIDTH} = \text{Throttle Low Pulse Width} + 100\mu\text{S}$

If the difference between Throttle High Pulse Width and Throttle Low Pulse Width (pulse delta) is less than 600 μS , the 100 μS threshold is reduced proportional to the difference in the pulse delta when the pulse delta is between 600 μS and 200 μS . For example, if the pulse delta is 400 μS , the $\text{RC_MOT_ON_PULSE_WIDTH}$ shall be $\text{Throttle Low Pulse Width} + 50\mu\text{S}$. If the pulse delta is below 200 μS , the GASSENSOR rejects the input, thus any REAL-TECT system must also reject that input.

At the conclusion of the last ENTER button press, the LCD shall clear and display:

"GASSENSOR Cal LO <cr> (ENTER) to Cont" and continuously output a 1.0 ms, 30 Hz pulse.

(This is a representative spare R/C receiver throttle low pulse and will be used by the GASSENSOR to determine its $\text{RC_MOT_ON_PULSE_WIDTH}$ in order to output the net logical Motor state (per Table 1 above) to the Base Station).

This condition shall persist for approximately 3 seconds after which LO on the display will change to HI and the output pulse width will change to 2.0 ms at 30 Hz to simulate R/C receiver throttle high pulse width.

The operators are required to press appropriate buttons on the GASSENSOR while the GDGT output to the GASSENSOR toggles between 1 ms and 2 ms every 3 seconds.

Once the GASSENSOR has accepted both inputs via audible beeps, the Operator shall press ENT on the GDGT to denote completion of the calibration and ready the system for operation/flight.

After final user input, the LCD shall clear and display

“NODES< Primary NODE ID>,<BACKUP NODE ID>,CH<RF CHANNEL>” on line 1 and

“RC:”<RC_MOT_STATUS>RF:”<RF_MOT_STATUS>WM:”<Watt Minutes> on line 2.

2.1.6 Functional Description: ATT

The ATT shall accumulate Watt-Minutes (WM) independently of any telemetry transmissions including during out-of-range conditions.

Power measurement shall be at a minimum rate of 10 Hz. The accuracy of the determined energy (WM) shall be within the range +/- 2%. This accuracy must be achieved across an input voltage range of 20 and 42 volts, a current range of 50 to 250 amps and an ambient temperature range of 0 to +50 degrees centigrade.

At Power On, the microprocessor shall read and store any current sensor tare value in flash memory and zero the Energy accumulation.

The first WM transmission is assumed to be a “Motor Test Blip” at less than 50 WM. After this Test Blip the ATT shall transmit “0000” for WM.

When the accumulation exceeds 50 WM the WM will accumulate and the value will be transmitted.

When Motor-On is detected, Motor-ON transmissions shall be nearly continuous at 30 Hz.

When the motor current subsequently decays below the threshold determined during calibration, the Motor-OFF event will be transmitted at 30 Hz.

When the Motor-OFF condition has persisted for 45 times at 30 Hz, the ATT shall transmit accumulated WM, alternating this with Motor-OFF at a repetition rate of 1 Hz until the next Motor-ON event is detected or power is interrupted.

The telemetry motor transmissions shall reset to Motor-OFF within 20 ms of motor power decaying to below 275 ± 50 Watts.

3 Verification Testing

3.1 Equipment required for testing

- a) Regulated power supply capable of 0-50 volts output.
- b) Current source capable of up to 250 amps.
- c) Constant current electronic load capable of up to 250 amps.
- d) Suitable meters and shunts for reading current and voltage to 1% or better accuracy.
- e) Stopwatch capable of at least .1 second resolution.
- f) “Servo tester” to generate a motor command pulse 900 to 2000 μ S.
- g) Oscilloscope.
- h) Complete carbon fibre F-5B aircraft including motor, batteries, speed control, R/C receiver, etc. (e.g. Avionic B14 model or equivalent) for RF range testing.
- i) 10 S LiPo Battery with 6 mm connectors.

3.2 Ground Display GASSENSOR Transceiver Tests

TEST 1	GDGT POR and Version Display.
TEST 2	GDGT Invalid NODE ID Rejection
TEST 3	GDGT Primary NODE ID Input.
TEST 4	GDGT Valid Backup NODE ID Input.
TEST 5	GDGT Invalid Backup NODE ID Rejection
TEST 6	GDGT RF Module Channel Adjustment
TEST 7	GDGT R/C Throttle Hi Input
TEST 8	GDGT R/C Throttle Lo Input
TEST 9	GDGT GASSENSOR HI/LO Calibration
TEST 10	GDGT Flight Ready Display
TEST 11	GDGT R/C Motor Status Toggle : 1/0/N
TEST 12	GDGT RF Motor Status Toggle : 1/0/N
TEST 13	GDGT WM Accumulation Display
TEST 14	GDGT Wrong Node RF Message Reject
TEST 15	GDGT Wrong Checksum RF Message Reject
TEST 16	GDGT RC/RF Motor Status Logic (All Table 1 Conditions)
TEST 17	GDGT RF Dropout Persistence- Motor ON: < 37.5 ms
TEST 18	GDGT RF Dropout Persistence-Motor OFF: < 1.5 S
TEST 19	GDGT RC/RF Motor Status Change without GASSENSOR interruption
TEST 20	GDGT Heartbeat Message Persistence
TEST 21	GDGT RF Motor ON Time Delay: < 20 ms
TEST 22	GDGT RF Motor OFF Time Delay: <20 ms
TEST 23	GDGT RF Motor ON/OFF Transition Persistence : $1 \pm .1$ S

3.3 ATT Tests

TEST 24	ATT Message Content: NODE ID=Labeled ID
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TEST 25	ATT Message Content: Checksum Accuracy
TEST 26	ATT Message Content: MOTOR-ON
TEST 27	ATT Message Content: Motor-Off
TEST 28	ATT Message Content: Watt-Minutes
TEST 29	ATT Message Content: 1 st Motor ON < 50 WM="0000"
TEST 30	ATT Motor ON/OFF Toggle Delay: < 20 ms
TEST 31	ATT Motor OFF/ON Toggle Delay: <20 ms
TEST 32	ATT WM Calibration Accuracy: $\pm 2\%$ over Temp and Current Range

3.4 Systems Test

TEST 33	RF RANGE : >300 metres, over flat unobstructed ground, with the ATT in a complete epoxy-carbon fibre based Model (such as an Avionic B14) and with the GDGT maintained 1 metre AGL.
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4.0 Competition Information

4.1 Installation Environment

- 4.1.1 The installation and functioning of the REAL-TECT ATT, in conjunction with the competitor's transmitter (including its programming), receiver and ESC, is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.
- 4.1.2 The REAL-TECT ATT must be installed in such a way that it is protected from mechanical damage.
- 4.1.3 The REAL-TECT ATT must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the current or voltage sensing at any time is prohibited.

5.0 Verification of REAL-TECT Components

This section details procedures for verifying the operation of Approved REAL-TECTs for use in the F5B competition environment. It is provided as a guide for competition organizers.

5.1 Purpose of Verification

All REAL-TECT Systems used in an F5B competition are required to be of an Approved Type. The verification process serves to ensure that no individual REAL-TECT system operates in a manner different from others in use at the competition.

5.2 ATT WM Calibration Testing

Testing will be done at a nominal 1500 Watts. The voltage for testing will be 30 volts and the current will be 50 amps.

5.3 Test Sequence for Checking WM Calibration (TEST 32)

- a) Connect a GDGT to a GASSENSOR and Servo Tester. Configure the NODE ID to match the ATT under test. Complete the GDGT setup sequence to the Operate Mode.
- b) A 1500 Watt simulated load will be connected to the ATT current sensor using the above test equipment. The ATT power supply shall be 42 vdc (equivalent to a 10 S LiPo).
- c) A Test Current blip shall be applied to ensure the associated GDGT display reads "0000".
- d) The 1500 Watt Load and the stopwatch will be started simultaneously.
- e) After a period of 70 seconds, both the load and the stop watch will be stopped simultaneously. GDGT WM shall display between 1715 and 1785 WM. <this is $\pm 2\%$ >.

5.4 Test Pass/Fail limits

REAL-TECT Systems and Systems components will considered to have passed if all the Test Requirement of Section 5.2 and 5.4 are met.

5.5 Firmware Revision Level

- 5.5.1 All REAL-TEST ATTs will have an anti-tamper sticker showing the version of firmware.
- 5.5.2 All GDGTs will display firmware revision upon start up.

6.0 Submission of REAL-TECT for Approval

The CIAM is establishing a Working Group to assume responsibility for the approval process. The services of a technical expert may be used to undertake the practical testing on behalf of the Working Group. Further details will be advised in due course, together with the specific requirements concerning the presentation of devices for approval.

Devices submitted for approval must be to normal production standard and must incorporate firmware at the revision level that is to be evaluated.

Additional supporting information may be required at the request of the technical expert undertaking the evaluation on behalf of the CIAM Working Group.

Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.

6.1 Withdrawal of Approved Status

An approved F5BEL/F5BELo may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the REAL-TECT.

If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the REAL-TECT, the validity of all REAL-TECT on the Approved List will be subject to review.

Section 4B – Performance Requirements Definition for F5B and F5F Electronic Energy Limiter and Logger Devices

Initial Publication by N. Hübner: 20 Nov, 2015

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1.2	Philosophy behind the F5BELIM/F5BELOG Specification
2	Information for Manufactures
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1.1 Purpose of this Document

The FAI CIAM rule making/amendment process is geared to maintain competition rule stability and as such presents some difficulties in areas subject to rapidly changing technologies. To overcome this problem, CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” Working Group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of F5B Energy Limiters / Loggers (F5BELIM/F5BELOG) for use in F5B or F5F competitions. The information contained herein is relevant for F5BLIM/F5BLOG manufacturers, competition organizers, competitors and persons involved in the evaluation of F5BELIM/F5BELOG.

1.2 Philosophy behind the F5BELIM/F5BELOG Specification

As long as the F5B REAL-TECT System is not used worldwide for any F5B or F5F competition an additional specification for the up to now used F5BELIM/F5BLOG systems is required.

Especially international F5B or F5F competitions attract large numbers of competitors, each of whom is entitled to two models. Consequently, completion organizers could expect to be presented with a significant number of F5BELIM/F5BLOG from a number of different manufacturers for scrutinizing and verification purposes, pending on the decision of the organizer if F5BELIM or F5BELOG have to be used during the whole competition. It is against this background that the F5BELIM/F5BELOG has been formulated with the following objectives:

- a) To accurately determine the consumed energy as required for the competition.
- b) To ensure that the F5BELIM/F5BELOG can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- c) To provide a standard method of operating so that, irrespective of F5BELIM/F5BELOG manufacture, verification testing can be performed easily with a defined test procedure.
- d) To define standard connectors for compatibility with test facilities and to reduce risk of accidental damage during handling by competition officials.
- e) To provide a standard method of determining that the F5BELIM/F5BELOG is of the type for which an approval has been granted.

2 Information for manufacturers

2.1 Technical Specification

This section is the formal specification for F5BELIM/F5BELOG intended for use in the F5B or F5F competition environment.

An F5BELIM/F5BELOG carried in an F5B model shall fulfill the following technical specifications:

- a) Current measurement will be by use of resistive shunt resistor or current sensing Hall sensor type device installed in the positive side of the power lead between the battery and the motor controller.
- b) Voltage measurement will be made between the positive and negative leads of the battery that supplies power for the motor. The negative pole has to be installed at the controller side.
- c) The method of converting current and voltage to energy units (watt-minutes or Wmin) must be such that the specified accuracy is ensured over the operating range. The minimum nominal current the shunt or Hall sensor has to handle, depending on the nominal voltage is:

	21.0VDC (5S LiPo)	-> 400A
	42.0VDC (10S LiPo)	-> 240A
for F5F only:	16.8VDC (4S LiPo)	-> 400A
	25.2VDC (6S LiPo)	-> 300A

For different nominal voltages the minimum required current has to be interpolated.

- d) F5BELIM only: The motor will be stopped once the energy limit has been reached. The motor cut off will continue till landing of the model after the duration task. Motor cut off has to be at 1.2ms motor signal.
- e) The F5BELIM only must have compatibility with ESCs using the nominal 1 to 2 milliseconds control signal from the receiver such that no low-speed motor drive occurs when "motor stop" is instituted either by the competitor or the F5BELIM
- f) The supply of power to the F5BELIM/F5BELOG shall be that available from the receiver output to the ESC at the receiver motor command connector. If the power source to the receiver is supplied via the ESC motor command connector, it is permissible to be supplied to the receiver via the F5BELIM/F5BELOG.
- g) The connectors of the F5DELIM/F5BELOG shall be of a universal type compatible with JR/Futaba and must be proof against any misalignment that could result in electrical damage to the F5BELIM/F5BELOG or connected equipment. Where an extension lead is used to satisfy this requirement, it must be secured to the F5BELIM/F5BELOG in such a manner that it cannot be unintentionally removed.
- h) F5BELOG only: the consumed energy during the flight will be registered after landing with a monitor capable to be connected to the lead of the F5BELOG. This lead of the F5BELOG must be accessible from outside of the model downstream of the wing trailing edge for safety reasons. For planes with pusher prop upstream of the leading edge.
- i) The connector to show the used energy level stored in F5BELOG to the display should be of a type as small as possible however suitable for multiple connection and the same for all F5BELOG from all manufacturers.
- j) The monitor to register the consumed energy has to show the used software version, the adjusted settings and the consumed energy.
- k) The F5BELOG has to store the flight data with a time stamp in non-volatile memory.
- l) The F5BELIM/F5BELOG start data recording when the receiver is switched on.

2.2 Characteristics of the F5BELIM/F5BLOG

2.2.1 The accuracy of the determined energy (wattminutes or Wmin) shall be within the range of +/- 2% from across 800 Wmin up to 2200 Wmin within a temperature range of 0 to +50 deg centigrade at an operating voltage of 12.6VDC-42VDC (3 to 10 LiPo).

2.2.2 The calculation of the energy shall incorporate any calibration parameters provided by the current sensor manufacturer.

2.2.3 The sampling rate of the power management used for energy usage determination shall be a minimum of 10 samples per second.

2.2.4 The energy created by a power level below 25W shall not be taken into account.

2.2.5 There must be no configurable settings or adjustments that allow the user to modify the operation of the F5BELIM/F5BELOG such that it will operate outside the requirements of the competition rules.

2.3 Starting Condition

The F5BELIM/F5BELOG shall initialize using following procedure:

Immediately on application of power to the F5BELIM/F5BELOG, means battery is connected to the ESC or receiver is switched on, it will zero out the energy accumulator and start to accumulate the energy and store it in non-volatile memory.

3 TECHNICAL CONSIDERATIONS

This section is provided as design guidance on F5BELIM/F5BELOG intended for use in F5B (F5F) competition environment.

3.1 Restriction on User Configuration Options

Whilst for commercial considerations manufacturers may wish to produce F5BELIM/F5BELOG adaptable for differing applications, when used in F5B (F5F) competitive environment, the installed firmware shall have dedicated F5B (F5F) ELIM/ELOG functionality with no user configurable options. The purpose of this is to eliminate any possibility of errors in operation due to incorrect configuration settings.

3.2 Connector Compatibility

The F5BELIM/F5BELOG will use standard JR/Futaba type connectors for both the signal input and outputs. The power connectors will be 6mm connectors. The power connector that connects to the + battery will be the male type on the F5BELIM/F5BELOG and the connector that connects + to the motor controller will be the female type. The connector to the energy registration display must be a 4-pole female JR type connector.

3.3 ESC Compatibility

There is no defined standard for the manner in which ESCs (electronic speed controllers) initialize to the range of motor command signals. In addition, programmable transmitters allow the user to set the motor command signals to any value within the programmable range.

3.3.1 The F5BELIM/F5BELOG specification of motor on threshold at 1.2 milliseconds has been selected as being compatible with most known equipment but cannot be a guarantee of correct operation for every combination of transmitter, receiver and ESC.

4 COMPETITION INFORMATION

4.1 The installation and functioning of the F5BELIM/F5BELOG, the transmitter and its programming, receiver and ESC, is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.

4.2 The F5BELIM/F5BELOG must be installed in such a way that it is protected from mechanical damage.

4.3 The F5BELIM/F5BELOG must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the current or voltage sensing at any time is prohibited.

4.4 For F5BELIM only the ESC must always operate via its series connection to the F5BELIM and not with a direct connection to the receiver.

4.5 Any manipulation or soldering at the F5BELIM/F5BELOG devices is prohibited.

Exception: it is allowed to adjust the length of the cable connecting the F5BELOG to the monitor for energy registration.

4.6 If other connectors for energy registration are used, the competitor has to offer a suitable adapter.

Attention: for a penalty of unreadable information from the logger/limiter see appropriate section in F5B rules.

5 Verification of F5BELIM/F5BELOG

This section details procedures for verifying the operation of Approved F5BELIM/F5BELOG for use in the F5B (F5F) competition environment. It is provided as a guide for competition organizers.

5.1 Purpose of Verification

All F5BELIM/F5BELOG used in an F5B (F5F) competition are required to be of an Approved Type. The verification process serves to ensure that no individual F5BELIM/F5BELOG operates in a manner different from others in use at the competition.

5.2 Verification

5.2.1 Testing condition

F5BELIM/F5BELOG as well as for F5F Calibration testing will be done at a nominal 1500 watts. The voltage for testing will be 30 volts and the current will be 50 amps for F5B. For F5F 15 volts and 100 amps.

5.2.2 Required Equipment for testing

- a) Regulated power supply capable of 0 – 30 Volts output
- b) Current source capable of up to 100 amps
- c) Electronic load capable of up to 30 volts and 50 amps = 1500 watt for F5B
- d) Electronic load capable of up to 15 volts and 100 amps = 1500 watt for F5F
- e) Suitable meters and shunts for reading current and voltage with 1% or better accuracy
- e) Stopwatch capable of at least 0.1 second resolution
- f) A “servo simulator” to generate a “motor” full on and full off command pulse (~1ms...2ms)
- g) Suitable meter or oscilloscope to monitor command output of F5BELIM/F5BELOG

5.2.3 Test sequence for checking WMin accuracy

- a) A 1500 watt simulated load will be applied to the F5BELIM/F5BELOG and (F5F as specified) using the above test equipment (Servo simulator)
- b) The stopwatch will be started when the “motor” command pulse is changed to more than 1.2 ms and stopped below 1.2 ms after 70 sec. The consumed energy after 70 sec. shall be between 1715 and 1785 WMin <this is $\pm 2\%$ >

5.2.4 Firmware Revision Level

All F5BELIM/F5BELOG will show the approved firmware signification in addition to the Wattminutes on the display when the energy is registered after landing.

5.2.5 Test Pass / Fail limits

A F5BELIM/F5BELOG will considered to have passed if systems and components have passed all the test requirement of sections 2.1, 2.2, 2.3, 3.1, 3.2, 3.3.3, 5.2.1, 5.2.2, 5.2.3 and 5.2.4 are met.

Section 5 – F1 Free Flight V1.3

Initial Publication: August 13 2019

By Ian Kaynes, CIAM F1 Subcommittee Chairman

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EF1 Altimeters to determine flight duration

Update Record

Produced by F1 Subcommittee Chairman and developed in Subcommittee and with Chairman EDIC through a number of drafts.

EF1.1 Introduction

The 2017 CIAM Plenary Meeting approved a proposal to have optional confirmation of flight time in flyoffs by review of altimeter records. This has been included in Volume F1 of the Sporting Code at paragraph F1.2.7.

At the time of approval by CIAM the proposal had not been considered by EDIC. The purpose of this document is to propose the minimum specification of altimeters suitable for the intent of that proposal but with improved control to facilitate the application of altimeters:

- a) identification of devices so that a known device is used for a specific flight without requiring the pattern recognition procedure mandated in F1.2.7
- b) assurance that the time basis of the altimeter records are accurate and cannot be subject to tampering
- c) to require a standard interface for replay of the altimeter records so that this can be carried out in a consistent manner

The nearest equivalent device considered by EDIC is the altimeter used in Space Models. This device detects the moment of launch by a change of altitude which would be suitable for F1B and F1C launched from the ground, but not for F1A models which are released from the towline at altitude. It is not essential to have a specific indication of the moment of launch, which can be found by inspection of the altitude trace, but identifying moment of launch would simplify interpretation of the trace.

Another feature of the space model altimeter is that it accepts only one flight record and can only be used for another flight when it has been read and reset. While this is suitable for the mandatory use of altimeters in all space model flights and provides security against some forms of cheating, it would be unduly restrictive for the optional use of altimeters under the current F1 regulations which may also involve multiple flyoff flights.

In addition to the immediate need to meet the needs of F1.2.7 it is also considered that in future the use of an altimeter could provide altitude data which would allow a determination of the winner in a flyoff restricted below the normal specified durations for flyoffs. This would provide an alternative to the present occurrence of “DT flyoffs” which are outside the 2019 Sporting Code but provide a means to conclude a flyoff when conditions do not allow a normal long duration flyoff. A future change to use altimeters in this application requires the specification of the accuracy of the altitude data given in this document but which is not essential for use under F1.2.7.

EF1.2 Altimeter Specification

- a) Use a barometric pressure measurement technique.
- b) The calculation of pressure shall include any calibration parameters and recommendations provided by the pressure transducer manufacture
- c) The conversion of pressure measurements to altitude must conform to the international standard atmosphere as defined in Manual of the ICAO Standard Atmosphere ICAO 7488/3.

As an example, from this reference the relationship of altitude to pressure in the standard atmosphere at 15C is given by

$$\text{Altitude} = K1 * [1 - (P/P_0)^{K2}] \text{ metres}$$

where $K1 = -T/L = 44330.769$ metres

$K2 = -R L/g_0 = 0.190263$

T is temperature Kelvin

L is lapse rate (deg per metre)

R specific gas constant

g_0 acceleration due to gravity

- d) The accuracy of the altitude measurement should be within 2 metres or 0.5% of altitude, whichever is the greater. This should be maintained over a temperature range of -10°C to +50°C with a working range of at least 1000 metres from sites at up to 2000 metres altitude.
- e) Altitude should be recorded in metres to at least one decimal place
- f) Timing accuracy should be within 1 sec over a time of 1000 sec.
- g) The data sampling rate shall be not less than 5 samples per second.
- h) Recording capacity must be sufficient for at 40 minutes recording.
- i) Be powered by connection to a battery included in the model or by a dedicated battery, either rechargeable or non-rechargeable, having sufficient capacity to operate the altimeter for a minimum of one hour before requiring recharging or replacement.
- j) Retain the flight data including after power removed.
- k) Be equipped with a data communication interface to be used for accessing data and the software and hardware required for the interface must be defined. It is desirable that a common standard be adopted, to be defined in a future revision of the specification after discussion with manufacturers.
- l) Carry a unique serial number allocated to the altimeter by the manufacturer and which is also accessible electronically with the flight data. This serial number may be marked on a durable label securely attached either directly to the altimeter or on leads from which it cannot be removed.
- m) Provide an audible and/or visual indication of its operating status. For the purposes of this document 'flashes' shall mean audible beeps and/or visual flashes.
- n) Have no configurable settings or adjustments that allow the user to modify its operation such that it will operate outside the requirements of the competition rules.
- o) To record multiple flight data sequentially so that the last flight made is shown by the last set of data shown on data playback.

EF1.3 Operation

Following the application of power, the altimeter shall determine whether it is in a Flight Recording Mode or connected to a readout device in Data Communication Mode.

EF1.3.1 Flight Recording Mode

- a) On application of power, the altimeter shall indicate readiness to accept flight data by single flashes with a repetition rate of once every second.
- b) Multiple flights must be recorded in sequential order so that last flight is identifiable.
- c) The altimeter shall continue recording until power is removed or has reached storage limits.

EF1.3.2 Data Communication Mode

Data Communication Mode is required for retrieval of flight data.

In Data Communication Mode the altimeter will indicate that it is connected in this mode and contains flight data by 2 flashes every second.

The altimeter must connect with a digital device such as a computer equipped with a software package that will:

- a) Display the firmware revision level of the processing firmware loaded onto the altimeter.
- b) Display the unique serial number of the altimeter.
- c) Display the full digital output of the altimeter's altitude versus time data in graphical and tabular format for the last flight recorded on the altimeter.
- d) Record a file containing the information in (a), (b) and (c).
- e) Permit resetting the altimeter so that any previously recorded flight data is deleted and can no longer be retrieved from the altimeter.

When making an official flight for which the competitor might require the altimeter record to be considered, the timekeeper will record the serial number of the altimeter installed in the model for that flight. This represents the evidence of which altimeter provides the evidence for that flight if required.

EF1.4 Device Approval

All altimeters used to provide flight time information in FAI competitions must be of an approved type. In order to give CIAM approval to a device, the verification procedure must be followed:

- a) Record the manufacturer, and the model and software variants of the device tested.
- b) Demonstrate that the recorded altitude has a consistent variation with actual altitude and meets the accuracy requirement. This requires the use of a partial vacuum chamber. To confirm accuracy a calibrated altimeter should be included in the chamber and for consistency of altimeters, a number should be tested simultaneously.

The procedure is:

- i) Place the altimeters in the chamber and switch on to start flight recording mode
- ii) Reduce pressure in the chamber as smoothly as possible until the pressure has dropped to 85% of the ambient pressure (which represents an altitude gain of 1350m), then continue to 70% pressure (2907m), then slowly restore the pressure to ambient.
- iii) Stop altimeter recording and inspect the altitude traces. It is required that the altitude should change consistently without any reversals of altitude while the pressure was decreasing and increasing. Checks should be made of the altimeter accuracy by the changes between altitude readings at ambient pressure and the 85% and 70% pressures.
- c) Check timing accuracy with the following procedure:
 - i) Switch on the altimeter to start flight recording mode
 - ii) Rapidly raise the altimeter at least 1m while starting a stopwatch
 - iii) Approximately 17 minutes later rapidly lower the altimeter while stopping the stopwatch.
 - iv) Download the data from the altimeter and check that the time indicated by the trace is not significantly more than one second different from the time recorded on the stopwatch.
- d) Check effect of temperature at ambient temperature
 - i) Switch on the altimeter to start flight recording mode
 - ii) Lower temperature in enclosure containing altimeter to -10°C
 - iii) Slowly raise the temperature to 50°C
 - iv) Stop altitude recording and inspect altitude traces. Variations of altitude over the temperature range should not exceed 2m

EF2 Energy Limiters for F1Q

Update Record

Produced by F1 Subcommittee Chairman and developed in Subcommittee and with Chairman EDIC through a number of drafts.

V1.3 includes removal of irrelevant mention of pressure sensors in EF2.11.

EF2.1 Introduction

When F1Q was first introduced the power limitation was by battery size and duration of motor run. In 2012 this was replaced by a concept of energy allowance which was observed by either power measurement and run duration or by an energy limiter. In 2018 the regulation was changed to require an energy limiter. As a mandated electronic device in the model specification, it is required that and EDIC specification and approval of devices is established.

An energy limiter for use in F1Q under these regulations will be referred to as an F1QEL.

One feature required by the specification is that the F1QEL has a display showing the energy limit that has been set. This is not available on all energy limiters currently used and it could be appropriate to delay the implementation of this part of the requirements to a later date than the other aspects of the specification.

EF2.2 Requirements

The energy allowed for an F1Q model flight is defined by the Sporting Code. In order to respect the limit it is necessary to ensure that the energy limiters used in F1Q are of adequate accuracy which cannot be adjusted by the competitor. It is necessary that the energy limiter operates according to the set energy limit and that this limit is displayed for easy confirmation by officials.

The F1QEL Specification has been formulated with the following objectives:

- a) To accurately limit the consumed energy as required for the competition.
- b) To ensure that the EL can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- c) To provide a standard method of operation so that, irrespective of F1QEL manufacture, verification testing can be performed easily with a defined test procedure.
- d) To display the value of the energy limit which has been set
- e) To define standard connectors for compatibility with test facilities and to reduce the risk of accidental damage during handling by competition officials.
- f) To provide a standard method of determining that the F1QEL is of the type for which an approval has been granted.
- g) To guarantee the particular version for which the device was approved.

EF2.3 Technical Specifications

This section is the formal technical specification for F1QEL.

- a) Current measurement will be by use of resistive shunt resistor or current sensing Hall effect sensor installed in the positive side of the power lead between the battery and the motor controller.
- b) Voltage measurement will be made between the positive and negative leads of the battery that supplies power for the motor. The battery voltage measurement is to be taken on the battery side of the measurement device.

- c) The method of converting current and voltage to energy units (joules), must be such that the specified accuracy is ensured over the operating range. The calculation will be $\text{joules} = \text{Watts} \times \text{seconds motor run}$, where $\text{Watts} = \text{volts} \times \text{amps}$.
- d) The motor will be stopped once the accumulated energy limit has been reached. The stop signal may be either by direct command of the motor through the ESC or by a stop signal sent to timer or other intermediate device which controls the ESC.
- e) The F1QEL must have compatibility with ESCs using the nominal 1 to 2 milliseconds (ms) control signal such that no low-speed motor drive occurs when “motor stop” is instituted either by the F1QEL or any other device.
- f) The supply of power to the F1QEL shall be that available from the ESC motor command connector or from the battery power lead. If the power source to the timer and other devices on the aircraft then it is permissible for this to be supplied via the F1QEL.
- g) The connectors of the F1QEL shall be of a universal type compatible with JR/Futaba and must be proof against any misalignment that could result in electrical damage to the F1QEL or connected equipment. An extension lead may be used to satisfy this requirement.
- h) Except for setting the maximum energy, there must be no configurable settings or adjustments that allow the user to modify the operation of the F1QEL such that it will operate outside the requirements of the competition rules.
- i) The F1QEL must start measuring energy used from the end of a start switch signal indicating the moment of launching the model

EF2.4 Accuracy of the F1QEL

- a) The accuracy of the determined energy (joules) shall be within the range $\pm 2\%$. This accuracy must be achieved across input voltage between 4 and 20 volts and across 2 to 20 amps and ambient temperature range of -10 to $+50$ degrees centigrade.
- b) The calculation of the energy shall incorporate any calibration parameters provided by the current sensor manufacturer.
- c) The sampling rate of the power measurement used for energy usage determination shall be a minimum of 10 samples per second.

EF2.5 Starting condition

The F1QEL shall initialize using the following procedure:

- a) Immediately on application of power to the F1QEL it will zero out the energy accumulator.
- b) For the first 5 seconds after application of power the F1QEL must display the firmware revision level.
- c) After 5 seconds the F1QEL must display energy consumed on the last run and the current energy setting.
- d) The motor command will initialize to the “off state” and will remain there until commanded otherwise.
- e) The limiter must have the capability to detect the moment of launching the model via a parallel connection to the start switch of the timer. After the start switch is pressed, the limiter must let pass the motor command signal coming from timer to ESC. The F1QEL must start to accumulate energy consumption (joules) from the time of switch release (when the model is released from the competitor’s hand).

EF2.6 Stopping condition

When the allowed energy amount is reached the F1QEL must generate a motor stop signal and/or set the duration of motor control pulses to 1 ms (motor stop condition).

EF2.7 Technical Considerations

This section is provided as design guidance on F1QELs.

EF2.7.1 Connector Compatibility

The F1QEL shall use standard JR/Futaba type connectors for both the signal in and outputs and the signal from the start switch.

The power connectors will be 3.5mm connectors with a female to accept the positive lead from the battery and a male plug to connect with the positive lead to the ESC. A connector from the F1QEL to the negative lead between the battery and ESC should be fitted with a male 2mm plug.

EF2.7.2 ESC Compatibility

F1QEL specification assumes that when direct control of ESC is used the motor stops at pulses of 1ms. When an ESC is programmable, it should be set to operate with the defined motor-on threshold.

EF2.8 Installation Environment

- a) The installation and functioning of the F1QEL, the ESC, the motor, and the timer and other devices is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.
- b) The F1QEL must be installed in such a way that it is protected from mechanical damage.
- c) The F1QEL must be placed inside the model in the state as supplied by the manufacturer.
- d) The use of any method that modifies the current or voltage sensing at any time is prohibited.
- e) In the case when the F1QEL controls the ESC, the ESC must always operate via its series connection to the F1QEL
- f) The F1QEL display of energy setting must be readily visible for checking before launch or after the flight.
- g) When the F1QEL controls motor stop via a timer, the competitor must be able to demonstrate in ground test that the motor stop is initiated at the same time as the F1QEL display shows reaching the energy limit.

EF2.9 Verification of F1QELs

All F1QELs used in an F1Q competition are required to be of an approved type. The process of verifying a type for approval is to confirm that the operation and accuracy are according to the specifications.

EF2.9.1 Equipment required for testing

- a) A regulated power supply capable of 0 to 12 volts output.
- b) A current source capable of up to 10 amps.
- c) A constant current electronic load capable of up to 10 amps.
- d) Suitable meters and shunts for reading current and voltage with 1% or better accuracy.
- e) Stopwatch capable of at least 0.1 second resolution.
- f) A “servo simulator” to generate a motor command pulse.
- g) A start switch connected to the EL to indicate the start of EL accumulation.
- h) Suitable meter or oscilloscope to monitor motor command output of F1QEL.

EF2.9.2 Test sequence

Verification testing will be carried out for two energy limits and loads:

- a) Energy limit 800 joules and 40 Watts. The voltage for testing will be 10 volts and the current will be 4 amps.
- b) Energy limit 1600 joules and 80 Watts. The voltage for testing will be 10 volts and the current will be 8 amps.

The following test sequence will be conducted with these two energy limits and load:

- c) Set the F1QEL energy limit
- d) Apply the simulated load of the appropriate Watts using the above test equipment.
- e) The stopwatch will be started when the start switch is released.
- f) The output from the F1QEL will be monitored. When the motor pulse changes to below 1.2ms or the stop signal is given then the stopwatch will be stopped and the 40/80 Watt load terminated. The time will be recorded.

EF2.9.3 Test Pass/Fail limits

An F1QEL is considered to have passed if it reduces the motor command to below 1.2ms it issues the stop signal within 20 seconds after the start of test. The tolerance is +/-2%, i.e. 19.6 to 20.4 seconds. The same tolerance of +/-2% applies to the displayed energy (e.g 784 to 816 joules for the nominal 800 Joule test). Note that this is a statement of accuracy and does not imply any allowance to exceed the maximum motor run allowed in the F1Q rules as an additional constraint as well as energy limits.

An F1QEL is considered to have passed if all the test requirements outlined in this document are satisfied.

EF2.10 Firmware Revision Level

Displaying the firmware revision level shall occur for the first 5 s following the application of power to the F1QEL. On completion of the 5 s period, the display shall continue to show the recorded energy consumption of the last flight and the actual setting of energy amount, until a new energy accumulating start signal is detected.

EF2.11 Submission of F1QEL Devices for Approval

The CIAM is establishing a Working Group to assume responsibility for the approval process. The services of a technical expert may be used to undertake the practical testing on behalf of the Working Group. Further details will be advised in due course, together with the specific requirements concerning the presentation of devices for approval.

Devices submitted for approval must be to normal production standard and must incorporate firmware at the revision level that is to be evaluated.

Additional supporting information may be required at the request of the technical expert undertaking the evaluation on behalf of the CIAM Working Group.

Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.

EF2.12 Withdrawal of Approved Status

An approved F1QEL may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the F1QEL.

If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the F1QEL, the validity of all F1QEL devices on the Approved List will be subject to review.

EF3 Flight time recording system

Update Record

Produced by F1 Subcommittee Chairman and developed in Subcommittee and with Chairman EDIC through a number of drafts.

EF3.1 Introduction

This section presents a VERY INITIAL DRAFT OF SUGGESTIONS for a possible system to record flight times and transmits this to a central receiver system. The intent of this is for review and consideration of fundamentals before finalizing the definition of devices to be approved and used under such a system. Comments on practicality and features would be welcome from all, but especially from potential system developers

EF3.2 Outline

In F5B a telemetry system is used to communicate flight events to a base station. Radio Control duration classes are also working on landing detection systems.

There are many potential advantages for free flight with an automatic flight timing method which transmits the time to a central monitor station, such as:

- a) Accurate
- b) Possible System

The current optional use of an altimeter to demonstrate flight time provides a starting point for an on-board recording system referred to here as an F1REC. Compared to a basic altimeter, additional features are required for an F1REC:

- a) A start of flight indication – by a switch released when launching F1B, F1C, F1E and F1Q or a switch released when the towline leaves an F1A model (hook release, acceleration sensor, etc.)
- b) A landing detection system. At the simplest level this could be my monitoring altitude until no change was detected for several seconds (which would then be subtracted to give the flight time. This could be augmented by GPS and detection of no change of position for a similar defined time. If further reference was required this could be provided by an accelerometer, again showing no movement for a defined time.
- c) The landing system detection could also operate alongside a monitor of the flight time so that timing ceased when the maximum was reached, but there could be advantages in recording when and where the model actually landed (such as after a long DT descent)
- d) An advantage of including GPS would be reference to a standard time base so that times of events could be transmitted as actual time of launch and time of landing, reducing dependence on the time base within the device. It would also remove the need for a competitor to have a separate GPS location system.
- e) For power models (including electric) a means of detecting the end of the motor run and for electric models the energy consumed as recorded by the energy limiter.
- f) Transmission of the flight data to the base station. This would be identified by the unique serial number of the recording device. Possibly a start of flight and end of flight transmission would be sufficient, but with the possibility of monitor reports during the flight. Each transmission would be repeated until acknowledgement was received from the base station. The base station could inform the F1REC of the maximum time for the flight when it acknowledged a start of flight message.
- g) Storage of the flight data for use in the event of a communication problem with transmitted information.

EF3.3 Operation

It is assumed that each competitor would be required to have his own F1REC. These would have to be known approved devices which gave no opportunity to modify their operation and each one identified by a unique serial number. The F1REC would be installed in the model with input from a start switch and a power supply.

When preparing for an official flight the timekeeper would confirm the serial number of the device corresponded to the competitor. The timekeeper would observe the launch and the flight and note any exceptional events, such as a collision or part detached from the model.

It could be possible for the competitor to have a hand-held device recording the data transmissions from the model. This would provide confirmation of the data and inform him of motor run and the final flight time. In addition, if the system includes GPS, this device could also serve as his GPS receiver for retrieval of the model.

The base station would receive the data transmitted from the model, detect an infringement (such an over-run, or launch after the end of round) or record a valid flight time for the competitor recorded as using that F1REC.

Detailed sequence of operation when using transmission to a base station and the protocol for handling different categories of communication could be:

- a) When preparing to make a flight the timekeeper checks the serial number of the F1REC.
- b) When power is applied to the F1REC it clears previous flight data from memory and makes a transmission type 1 to give its identity to the base station, which responds with acknowledgment 1, possibly including the time of the maximum for the flight. The F1REC gives a ready signal (one flash every second) when the acknowledgment has been received and it is ready for flight (if GPS is included when the location has been determined)
- c) During the flight the F1REC records data on time, altitude, location, acceleration data, F1Q energy consumption, F1C motor run (according to what is available)
- d) At periods during the flight the F1REC periodically (e.g. every 15 seconds?) makes a transmission type 2 to report the instantaneous data values to the base station, which responds with acknowledgement 2.
- e) At the end of the flight, the F1REC makes a transmission type 3 to give the end of flight signal to the bases station including altitude at time of maximum, flight duration, location, F1Q energy consumption, F1C motor run, the base unit responds with acknowledgement 3 and terminates collecting data from this F1REC.
- f) If the periodic type 2 transmissions have not been received for at least 2 consecutive periods or if no end of flight type 3 transmission is received, then the last data received will be taken as the provisional end of flight and the data stored on the F1REC will be checked when the aircraft has been retrieved.

Some modification would be required to have the F1REC operate as a flight recorder without transmission to base station. This could be handled by a switch or setup procedure so that the unit does not wait for acknowledgements from the base station.

EF3.4 Possible Extensions

The system would give several additional features, which could be invoked for additional measurements of performance or for flight limitations or aids to organizers.

The base station record of the GPS data for each model would give a clear indication of where models were landing during a round and would provide information for adjustment of the maximum for the next round, to avoid hazards or retrieval problems.

Altitude at the time of reaching the maximum duration could be used as an additional indication of performance. The total altitude at the end of each max could be an alternative means of deciding the result instead of a flyoff, or could be used as a first cut in selecting perhaps 10% of competitors for a flyoff. The flyoff could be for a moderate maximum time if required by the conditions and the field size, with altitude then used for any final determination of the winner. One result of inclusion of altitude in the

classification of performance would be that maximum could be set to a time suitable for the field and wind conditions, since there would not be the same pressure to have long maxes in order to avoid large flyoffs. Avoiding long maxes in this way would also reduce any consideration of models being timed far away and out of sight.

Section 6 – F3G Logger and Energy Limiter V 1.1

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1.0 Introduction

1.1 Purpose of this Document

The FAI CIAM rule making/amendment process is geared to maintain competition rule stability and as such presents some difficulties in areas subjected to rapidly changing technologies. To overcome this problem, CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” Working Group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of F3G Logger and Energy Limiters (F3GLELs) for use in F3G competitions. The information contained herein is relevant to F3GLELs manufacturers, competition organisers, competitors and persons involved in the evaluation of F3GLELs.

1.2 Philosophy behind the F3G Logger and Energy Limiter (F3GLEL) Specification

International competitions attract large numbers of competitors, each of whom is entitled to three models. Consequently, competition organisers could expect to be presented with a significant number of F3GLELs from a number of different manufacturers for scrutinizing and verification purposes.

The F3GLELs specifications have been formulated with the following objectives:

- a) To accurately determine and log the parameters as required for the competition and defined by the FAI F3G rules.
- b) To ensure that the F3GLEL can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- c) To provide a standard method of operation so that, irrespective of F3GLEL manufacture, verification testing can be performed with a defined test procedure.
- d) To define standard connectors for compatibility with test equipment and to reduce the risk of accidental damage during handling by competition officials.
- e) To provide a standard method of determining that the used F3GLEL is in compliance with approved devices.
- f) To guarantee the particular version (eg. firmware) is used for which the device was approved.

1.3 Naming Convention

As the device used in F3G is acting as a logger (eg. recording the starting height) and a limiter (stopping the motor when the energy limit is reached) at the same time the term **F3G Logger and Energy Limiter** with the abbreviation **F3GLEL** is used across this document.

The FAI sporting code, Volume F3 Radio Control Soaring Model Aircraft, section F3G, is using the term **F3G LOG** and abbreviation **LOG**.

Both terms and abbreviations are referencing the same device and requirements and can be used interchangeably.

2.0 INFORMATION FOR MANUFACTURERS

2.1 F3GLEL Technical Specification

This section is the formal specification for F3GLELs intended for use in the F3G competition environment.

2.2 Technical Specifications

An electronic F3GLEL carried in an F3G model shall fulfil the following technical specifications:

- a) The F3GLEL must have compatibility with ESCs using the nominal 1 to 2 milliseconds (ms) control signal from the receiver such that no low-speed motor drive occurs when “motor stop” is instituted either by the competitor or the F3GLEL.
- b) The supply of power to the F3GLEL shall be that available from the receiver output to the ESC at the receiver motor command connector. If the power source to the receiver is supplied via the ESC motor command connector, it is permissible for this to be supplied to the receiver via the F3GLEL.
- c) The connectors of the F3GLEL shall be of a universal type compatible with JR/Futaba and must be proof against any misalignment that could result in electrical damage to the F3GLEL or connected equipment. Where an extension lead is used to satisfy this requirement, it must be secured to the F3GLEL in such a manner that it cannot be unintentionally removed.
- d) The F3GLEL shall be provided with a digital display. The display may be an integral part of the F3GLEL, or a separate module that is either carried within the model or is detachable. If the display is not an integral part of the F3GLEL, the method of transferring the information to the display shall be immune to any form of corruption or modification from external sources.
- e) Current measurement will be by use of a resistive shunt resistor or current sensing Hall sensor type device installed in the positive side of the power lead between the battery and the motor controller.
- f) Voltage measurement will be made between the positive and negative leads of the battery that supplies power for the motor. The battery voltage measurement is to be taken on the battery side of the measurement device if a resistive shunt is used.
- g) The method of converting current and voltage to energy units (Wmin), must be such that the specified accuracy is ensured over the operating range. The calculation will be watts (volts x amps) x minutes motor run.
- h) The motor will be stopped once the accumulated total energy limit (Wmin) according to the current F3G rules has been reached by output of a signal to the ESC of less than 1.2 ms.
- i) The motor will be stopped once the motor run time according to the current F3G rules has been reached by output of a signal to the ESC of less than 1.2 ms.

2.3 Characteristics of the F3G Logger and Energy Limiter (F3GLEL)

- a) The accuracy of the determined energy (WMin), shall be within the range +/- 2%. This accuracy must be achieved across input voltage between 12 and 35 volts (Lipo 3s to 8s) and across 20 to 70 amps and ambient temperature range of +10 to +50 degrees centigrade.
- b) The calculation of the energy shall incorporate any calibration parameters provided by the current sensor manufacturer.

- c) The sampling rate of the power measurement used for energy usage determination shall be a minimum of 10 samples per second.
- d) The accuracy of the determined 'Start Height' shall be within the range plus or minus 2.5 meters. This accuracy must be achieved across an ambient ground level pressure range of 750 to 1050 hPa and across an ambient temperature range of +10 to +50 degrees centigrade.
- e) The resolution of the display when showing the calculated 'Start Height', shall be 0.1 meter, this to be displayed with the format "xxx.x". Leading zero characters may be blanked. (When using the final calculated 'Start Height' for scoring purposes, the reading shall be rounded down to the nearest full meter.)
- f) The timing accuracy used for the Motor Stop function and 'Start Height' determination shall be equal to or better than 0.1 seconds.
- g) On completion of the period ending 10 seconds after "motor stop", the recorded 'Start Height' shall be continuously transferred to the display with a minimum repetition rate of once per two seconds. This function shall commence automatically when the recording of the 'Start Height' has been completed.
- h) There must be no configurable settings or adjustments that allow the user to modify the operation of the F3GLEL such that it will operate outside the requirements of the competition rules.
- i) The ESC output signal of the F3GLEL must follow the Rx input signal unless the stop condition is met.
- j) The display shall show the firmware revision level of the F3GLEL.
- k) The F3GLEL should only allow a single start (signal > 1.2ms) of the motor by the pilot unless the F3GLEL has been manually reset.
- l) The F3GLEL should allow a manual reset of its values and function. For example this can be achieved by disconnecting the F3GLEL from any power source.

2.4 Starting condition

The F3GLEL shall initialize using the following procedure:

- (a) The motor command will initialize to the "off state" and will remain there until the radio control command commands otherwise.
- (b) When the motor start command exceeds 1.2 ms the F3GLEL will zero out the energy and start to accumulate Wmin.

Notes:

- 1) A Motor Start command is defined as any control signal exceeding 1.2 milliseconds.
- 2) When the limit is reached refer to 2.2 d) for information on operation.

2.5 Recording the 'Start Height'

'Start Height' Determination - 'Start Height' determination requires the F3GLEL to perform a peak (height) detection function for the time duration commencing at switch-on initialisation, until 10 seconds after the motor has been stopped.

The F3GLEL shall record the 'Start Height' in accordance with the following procedure:

- a) Immediately on application of power to the F3GLEL it must determine a zero height reference level from which any subsequent 'Start Height' will be determined. Any previously recorded 'Start Height' must NOT be erased at this point.
This function shall be carried out with the model not higher than 2 meters above the ground level at the location specified within the competition rules.
- b) After power has been applied to the F3GLEL, the 'Start Height' determination and the start of the "Motor Run Timer" will be initiated by the Motor Start signal instituted by the competitor. Such initialisation must clear any previously recorded 'Start Height'.
- c) The cleared status of any previously recorded value shall be indicated on the display as a sequence of dashes (-----). This shall be carried out in such a manner that if a detachable display were to be connected it would show the cleared status.
- d) When the Motor Start signal has been instituted by the competitor, the F3GLEL will determine, and record, a new 'Start Height' replacing the previously cleared status.
- e) When power is removed from the F3GLEL, the 'Start Height' or cleared status indication must be retained by the F3GLEL. The application of power to the F3GLEL must never erase either the recorded 'Start Height' or the cleared status indication. Only the Motor Start signal instituted by the competitor may do this.

2.6 Motor Run Timer

The Motor Run Timer function must operate with the accuracy specified in item 2.3.

- a) On initial power-up of the F3GLEL, the Motor Run Timer must be protected from false triggering. After this period, the Motor Run Timer shall commence operation and 'Start Height' determination shall be initiated on the first occasion that the command signal exceeds 1.2 milliseconds.
- b) After the Motor Run Timer has been triggered, the timing of the motor run ceases when the command signal reduces below 1.2 milliseconds
- c) The F3GLEL has to automatically apply a defined motor stop command - signal below 1.2ms - to the ESC immediately when the Motor Run Timer reaches 30 seconds.
- d) The 'Start Height' determination process shall continue for a further 10 seconds after the instant of "motor stop".
- e) Once the definite stop command has been applied to the ESC, the system must not respond to any further changes in the motor command signal.
- f) When power is removed, the 'Motor Run Timer' must be retained by the F3GLEL. The application of power to the F3GLEL must never erase the recorded 'Motor Run Timer'. Only the Motor Start signal instituted by the competitor may do this.

2.7 Motor Energy Recording

The Motor Energy Recording function must operate with the accuracy specified in item 2.3.

- a) The Motor Energy Recording shall be initiated on the first occasion that the motor/ESC command signal exceeds 1.2 milliseconds. The Motor Energy has to be set to 0 (zero) at this occasion

- b) After the Motor Energy Recording has been triggered, the recording of the motor energy ceases when the motor/ESC command signal reduces below 1.2 milliseconds.
- c) The F3GLEL has to automatically apply a defined motor stop command - signal below 1.2ms - to the ESC immediately when the Motor Energy reaches the maximum defined value according to the F3G rules (350Wmin).
- d) Once the definite stop command has been applied to the ESC, the system must not respond to any further changes in the motor command signal.
- e) When power is removed, the 'Motor Energy' must be retained by the F3GLEL. The application of power to the F3GLEL must never erase the recorded 'Motor Energy'. Only the Motor Start signal instituted by the competitor may do this.

2.8 Retention of Motor Energy, Motor Run Time and Start Height

In the normal competition environment, the Motor Energy, Motor Run Time and Start Height reading is obtained immediately following the landing of the aircraft, without power removal from the system.

- a) There is a requirement for the F3GLEL to retain these parameters when power is removed. The primary reason for this is for purposes of reading dispute resolution and possible F3GLEL investigation away from the aircraft installation.
- b) Erasure of the stored 'Start Height' reading shall only occur in accordance with the sequence detailed in Items 2.4 (a) – (g).

3.0 TECHNICAL CONSIDERATIONS

This section is provided as design guidance on F3GLELs intended for use in the F3G competition environment.

3.1 Restriction on User Configuration Options

Whilst for commercial considerations manufacturers may wish to produce F3GLELs adaptable for differing applications, when used in the F3G competitive environment, the installed firmware shall have dedicated F3G functionality with no user configurable options. The purpose of this is to eliminate any possibility of errors in operation due to incorrect configuration settings.

3.2 Connector Compatibility

The F3GLEL shall use standard JR/Futaba type connectors for both the signal (receiver) in and outputs (ESC).

The power connectors will be 4mm connectors with a female on the positive lead from the battery and a male on the negative lead from the battery as well as a male on the positive side from the ESC and female from the negative side of the ESC.

The competitor has to provide suitable adapters if other connectors are used. These adapters need to be as short as possible, not adding any substantial resistance or influence the test of a competitor's F3GLEL device.

3.3 Electronic Speed Controller (ESC) Compatibility

There is no defined standard for the manner in which ESCs (electronic speed controllers) initialize to the range of motor command signals. In addition, programmable transmitters allow the user to set the motor command signals to any value within the programmable range.

The F3GLEL specification of motor-On threshold at 1.2 milliseconds (ms) has been selected as being compatible with most known equipment but cannot be a guarantee of correct operation for every combination of transmitter, receiver and ESC.

4.0 COMPETITION INFORMATION

4.1 Installation Environment

- a) The installation and functioning of the F3GLEL, the transmitter and its programming, receiver and ESC, is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.
- b) The F3GLEL must be installed in such a way that it is protected from mechanical damage.
- c) The F3GLEL must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the current-, voltage-, height-sensing or timer accuracy at any time is prohibited.
- d) Any manipulation or soldering at the F3GLEL devices is prohibited. Exception: it is allowed to adjust the length of the cable connecting the F3GLEL to the battery or ESC.
- e) The ESC must always operate via its series connection to the F3GLEL and not with a direct connection to the receiver.
- f) The F3GLEL shall be provided with a digital display. The display may be an integral part of the F3GLEL, or a separate module that is either carried within the model or is detachable.
- g) The F3GLEL must be installed in such a way that it is protected from pressure fluctuations other than changes in atmospheric pressure which result from the height of the model above ground level.
- h) The F3GLEL must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the true barometric pressure at any time is prohibited.
- i) Adequate venting shall be provided to minimize any uneven pressure distribution within the model in the region where the F3GLEL is installed.
- j) The competitor's F3GLEL shall be installed in such a manner that it can be easily removed from the model for verification purposes at any stage during the period of the contest.
- k) The ESC shall be configured that the motor power will not exceed 100 Watts for input signals below 1.2 ms.

4.2 Firmware Revision Level

All F3GLELs have to show the installed firmware version at the integrated or attached display. The firmware version used needs to be compliant with the version used for homologation.

5.0 VERIFICATION OF F3GLELS

This section details procedures for verifying the operation of Approved F3GLELS for use in the F3G competition environment. It is provided as a guide for competition organizers and manufacturers.

5.1 Purpose of Verification

All F3GLELS used in an F3G competition are required to be of an approved Type. The verification process serves to ensure that no individual F3GLEL operates in a manner different from others in use at the competition.

5.2 Testing Condition

- a) F3GLEL will be tested at a nominal 800 watts. The voltage for testing will be between 12 volts (3S) and 35 volts (8S).
- b) For overall validation and homologation of a F3GLEL device or firmware, the test sequences outlined below should be successfully performed and past with at least 3 independent F3GLEL devices.

5.3 Equipment required for testing

- a) A power supply capable of 12 to 35 volts output.
- b) A current source capable of up to 70 amps.
- c) A load capable of up to 800W / 70 amps.
- d) Suitable meters and shunts for reading current and voltage with 1% or better accuracy.
- e) Stopwatch capable of at least 0.1 second resolution.
- f) A “servo simulator” to generate a motor command pulse.
- g) Suitable meter or oscilloscope to monitor motor command output of F3GLEL.

5.4 Required Test Sequences

To test compliance with the F3G rules the following test sequences have to be performed. The F3GLEL needs to comply/pass all these tests successfully within the defined tolerances to be used in F3G competitions.

- a) Automatic stop of the ESC after 30 seconds by the F3GLEL without load (5.3)
- b) Automatic stop of the ESC after 30 seconds by the F3GLEL with load (5.3)
- c) Automatic stop of the ESC when energy limit (350Wmin) has been reached (5.3)
- d) Measurement of maximum achieved starting altitude (“F5J height”) with automatic (30 second) or manual stop of the ESC (motor)

An F3GLEL is considered to have passed if all the test requirements outlined in this document are satisfied.

5.5 Test sequence for checking timer accuracy without load

- a) A start signal (2.0 ms) will be issued to the F3GLEL
- b) The stop watch will start when the output signal (ESC side) of the F3GLEL reaches or exceeds 1.2ms
- c) The time is stopped when the output signal (ESC side) of the F3GLEL drops below 1.2 ms

The test is considered passed when:

- A. The F3GLEL output signal (ESC side) drops below 1.2ms within the time specified in the current F3G rules (30s + 0.1s)
- B. The displayed motor run time on the F3GLEL is within a 2% tolerance of the measured time.

5.6 Test sequence for checking timer accuracy with load

- a) An approximate 300 Watt simulated load will be applied to the F3GLEL using the above test equipment.
- b) A start signal (2.0ms) will be issued to the F3GLEL
- c) The stop watch will start when the output signal (ESC side) of the F3GLEL reaches or exceeds 1.2ms
- d) The time is stopped when the output signal (ESC side) of the F3GLEL drops below 1.2 ms and the load is terminated

The test is considered passed when:

- A. The F3GLEL output signal (ESC side) drops to 1.2ms within the time specified in the current F3G rules (30s + 0.1s)
- B. The displayed motor run time on the F3GLEL is within a 2% tolerance of the measured time.

5.7 Test sequence for checking energy limit accuracy

- a) A simulated load (800 watt) will be applied to the F3GLEL using the above test equipment.
- b) A start signal (2.0ms) will be issued to the F3GLEL. The load will be activated/started together with the start signal.
- c) The stop watch and energy recording (voltage & current) will be started when the output signal (ESC side) of the F3GLEL reaches or exceeds 1.2ms
- d) The motor control pulse of the F3GLEL will be monitored and the stopwatch and energy recording will be stopped when it drops below 1.2 ms. The load will be terminated at the same time.

The test is considered passed when:

- a) The F3GLEL output signal (ESC side) drops to 1.2ms with a energy recording of the test device within the specification of the current F3G rules (350 + 1 Wmin)
- b) The displayed start energy on the F3GLEL is within a 2% tolerance of the energy recording.

6.0 Submission of F3GLEL Devices for Approval

The CIAM is establishing a Working Group to assume responsibility for the approval process. The services of a technical expert may be used to undertake the practical testing on behalf of the Working Group. Further details will be advised in due course, together with the specific requirements concerning the presentation of devices for approval.

Devices submitted for approval must be to normal production standard and must incorporate firmware at the revision level that is to be evaluated.

Supporting documentation shall include the sensor manufacturer's data sheet on the pressure sensor device with particular reference to long term drift and temperature variation characteristics. Additional supporting information may be required at the request of the technical expert undertaking the evaluation on behalf of the CIAM Working Group.

Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.

6.1 Withdrawal of Approved Status

An approved F3GLEL may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the F3GLEL.

If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the F3GLEL, the validity of all F3GLEL devices on the Approved List will be subject to review.

Section 7 – F5L Time and Height Limiter

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1.0 INTRODUCTION

1.1 Purpose of this Document

The FAI CIAM rule making/amendment process is geared to maintain competition rule stability and as such presents some difficulties in areas subjected to rapidly changing technologies. To overcome this problem, CIAM Bureau instituted an “Electronic Devices in Competition (EDIC)” Working Group which will provide a system of “technical considerations” outside the rule making process that will enhance a class but not impinge on the class rules.

As part of that system, this document provides technical guidance on the specification, testing and application of Altitude Limiter / Motor Run Timer (ALMRTs) for use in F5L competitions. The information contained is relevant to ALMRT manufacturers, competition organizers, competitors and persons involved in the evaluation of devices for F5L.

1.2 Philosophy behind the ALMRT Specification

International competitions attract large numbers of competitors, each of whom is entitled to three models. Consequently, competition organizers could expect to be presented with a significant number of ALMRTs from a number of different manufacturers for scrutineering and verification purposes.

It is against this background that the ALMRT Specification has been formulated with the following objectives:

- a) To accurately determine the model height above ground as required for the competition.
- b) To accurately cut-off the motor at the altitude as required for the competition.
- c) To ensure that the ALMRT can only function as required by the rules of the competition and cannot be operated, unintentionally or otherwise, in any other manner.
- d) To provide a standard method of operation so that, irrespective of ALMRT manufacture, verification testing can be performed easily with a defined test procedure.
- e) To define standard connectors for compatibility with test facilities and to reduce the risk of accidental damage during handling by competition officials.
- f) To provide a standard method of determining that the ALMRT is of the type for which an approval has been granted.

2.0 MANUFACTURING SPECIFICATION

This section is the formal specification for ALMRTs intended for use in the F5L competition environment.

2.1 Altimeter – Motor Run Timer Technical Specification

The essential requirements of an electronic altimeter limiter and motor run timer carried in a F5L model are that it must:

- a) Use a barometric pressure measurement technique.
- b) For a single flight, measure the difference between the altitude at each moment of sampling after launch and the altitude of the point from which it was launched.
- c) The motor run timer must stop the motor 30 seconds after it is started at launch, if it has not already been stopped by the competitor within this limit or the model altitude has passed 90m.
- d) Have no configurable settings or adjustments that allow the user to modify its operation such that it will operate outside the requirements of the competition rules.
- e) To provide a standard method of operation so that, irrespective of ALMRT manufacture, verification testing can be performed easily with a defined test procedure.
- f) To define standard connectors for compatibility with test facilities and to reduce the risk of accidental damage during handling by competition officials.
- g) To provide a standard method of determining that the ALMRT is of the type for which an approval has been granted.

2.1.1 Barometric Pressure Measurement

- a) The conversion of pressure measurements to height information must be based on the International Standard Atmosphere, as defined in ICAO Document 7488/2. The detailed requirements are specified in Section 2.5 (Altitude Calculation) of this document.
- b) The method of converting pressure measurements to height information performed by the altimeter firmware must be such that the specified accuracy is maintained across the full working range.
- c) Within the altimeter system, it must have a smoothing or filtering technique that enables it to recognize and reject any abrupt pressure transients that are the result of flight dynamic events such as staging, activation of the motor, or wind gusts.
- d) It must have a measurement accuracy of better than 2 meters.
 - 1) This accuracy must be achieved across an ambient ground level pressure range of 750 to 1050 hPa and across an ambient ground level temperature range of 0 to +50 degrees Celsius.
 - 2) This accuracy shall include the effects of the manufacturer-specified accuracy of the pressure sensor. Such effects include:
 - The resolution of the analogue to digital conversion
 - Signal noise
 - Thermal drift across a period of two hours of operation when operated at 15 degrees Celsius
 - Hysteresis effects over the specified working range of the altimeter
- e) The calculation of pressure shall incorporate any calibration parameters provided by the pressure sensor manufacturer.
- f) Temperature measurements made by the pressure sensor device must not be used in the calculation of pressure to altitude, except in accordance with the sensor manufacturer's recommendations for sensor device compensation purposes.

2.1.2 Motor Run Timer

- a) The ALMRT motor run timer must not allow restart of the motor at any time during the flight.
- b) The ALMRT must have compatibility with ESCs using the nominal 1.0 to 2.0 milliseconds control signal from the receiver such that no low-speed motor drive occurs when "motor-stop" is instituted either by the competitor or the ALMRT.
- c) The supply of power to the ALMRT shall be that available from the receiver output to the ESC at the receiver motor command connector. If the power source to the receiver is supplied via the ESC motor command connector, it is permissible for this to be supplied to the receiver via the ALMRT.
- d) The connectors of the ALMRT shall be of a universal type compatible with JR/Futaba and must be proof against any misalignment that could result in electrical damage to the ALMRT or

connected equipment. Where an extension lead is used to satisfy this requirement, it must be secured to the ALMRT in such a manner that it cannot be unintentionally removed.

- e) The ALMRT may be provided with a digital display. The display may be an integral part of the ALMRT, or a separate module that is either carried within the model or is detachable. A detachable ALMRT display may be used to service any number of ALMRTs of the same approved type.

2.3 Characteristics of the ALMRT

- 2.3.1 The accuracy of the determined 'Motor-Stop Height' shall be within the range plus or minus 2.5 meters. This accuracy must be achieved across an ambient ground level pressure range of 750 to 1050 hPa and across an ambient temperature range of 0 to +50 degrees Celsius.
- 2.3.2 The calculation of 'Motor-Stop Height' shall incorporate any calibration parameters provided by the pressure sensor manufacturer.
- 2.3.3 The sampling rate of the pressure measurement used for 'Motor-Stop Height' determination shall be a minimum of 10 samples per second.
- 2.3.4 Any firmware processing of pressure measurements (or height information derived from such measurements) used for 'Motor-Stop Height' determination, must not result in a modification of the maximum height as would be detected by the single measurement sample representing the peak value.
- 2.3.5 The timing accuracy used for the 'Motor-Stop' function determination shall be equal to or better than 0.1 seconds.
- 2.3.6 Any additional features incorporated into the ALMRT by the manufacturer, must not cause any variation in the sampling rate or timing functions, nor affect the ability of the ALMRT to satisfy the requirements of the competition rules.
- 2.3.7 The display if present shall show the firmware revision level of the ALMRT with any leading zero characters blanked. Otherwise, the firmware revision shall permanently be printed on the device.
Digital display of firmware revision level shall occur at least for the first 3 seconds following the application of power to the ALMRT. On completion of the 3 seconds period, the display shall show the armed state.

2.4 Altitude Calculation

Pressure at a given location is caused by the weight of the atmosphere above. This varies with air density, humidity, gravity and temperature. In addition, temperature itself reduces as height increases. Consequently, some form of standardized definition is required to enable instrumentation based on barometric pressure to be used to determine height accurately.

The International Standard Atmosphere (ISA) assumes a mean sea level pressure of 1013.25 hPa at a temperature of 15 degrees Celsius, a specified temperature lapse rate with height, a specified gravitational force and a given molecular density of air.

Conditions on any day are unlikely to correspond exactly with the ISA model; a typical spread of 50hPa can occur at a given location due to normal meteorological conditions. Consequently, significant variations will occur from true geographic height. However, for aviation applications, including F5L, all instruments calibrated and operating in accordance with the ISA standard, will produce the same result.

- 2.4.1 In the lower atmosphere (up to 12km), the 15 degree Celsius ISA relationship for a pressure P, can be described by the following formula:

ISA Altitude = $K1 \times \{1 - (P/Po)^{K2}\}$ metres where

K1 = 44330.76923 metres

K2 = 0.190266669

Po = 1013.25 hPa (ISA sea level pressure)

K1 is derived from the formula:

$K1 = T/L$

Where T = temperature in degrees Kelvin

(15 degree Celsius is equivalent to 288.15 degrees Kelvin)

L = Lapse Rate 0.0065 degrees K per meter.

K2 is derived from the universal gas constant, temperature lapse rate, gravitational force, molecular mass of dry air and temperature with respect to absolute zero.

This formula results in a relationship between pressure and height that is represented by a curve that is steeper at high pressure/low altitude and less steep at low pressure/high altitude.

Note: In this context, results from calculations using the formula of the 1976 US Standard Atmosphere, agree closely with those of the ISA formula in the lower atmosphere and are considered acceptable over the F5L working range.

- 2.4.2 In order to achieve sufficient accuracy for F5L competition, the method of firmware processing of pressure data is important. Whilst ideally the ISA computation should be performed by an arithmetic calculation with maximum precision, processing constraints may make this impractical. It is permitted to use an algorithm approach to emulate the computation provided it can be demonstrated that an appropriate accuracy is maintained and no discontinuities are introduced in the ISA curve.
- 2.4.3 ALMRT manufacturers may wish to adopt pressure sensor modules that have an inbuilt pressure-to-height conversion. In such a case, it must be demonstrated that the computation complies with the accuracy requirements over the specified working range of the F5L application.
- 2.4.4 Pressure sensor offset errors will influence the 'Motor-Stop Height' calculation. However, since 'Motor-Stop Height' is the difference between two readings, much of any offset error is removed from the result.
- 2.4.5 In order to maintain overall ALMRT accuracy within the specified tolerance, the accuracy of the firmware ISA computation should be within the range plus or minus 0.5 metres.
- 2.4.6 During the launch phase, the dynamics of the flight envelope are subject to rapid change and the transition through the peak can occur within a short time period. Item 2.3.3 defines the minimum acceptable sampling rate for detecting the peak. It is not permissible for additional filtering or smoothing to be incorporated in the firmware as this can result in a downgrading of the ability to accurately detect the peak value.
- 2.4.7 Oversampling functions incorporated within the pressure sensor device may be used subject to being configured in a manner that maintains the integrity of readings taken at the specified sampling rate.
- 2.4.8 Pressure sensor devices:
- 2.4.8.1 Are subject to variation due to temperature. Many sensor manufacturers specify the inclusion of temperature data in the processing of the pressure data. Manufacturers' recommended firmware procedures must be incorporated into the ALMRT firmware.
- 2.4.8.2 The sensor can be affected by strong light falling on the sensitive area, therefore they must be mounted within the ALMRT in such a manner that the sensitive area is shielded from strong light levels.
- 2.4.9 The accuracy tolerance stated in item 2.3.1 is the maximum permitted when all possible variations are at worst case conditions. No individual factor should contribute a significant proportion of the total permitted variation.
- 2.5 Determination of the 'Motor-Stop Height'
- 'Motor-Stop Height' determination requires the ALMRT to perform a peak (height) detection function for the time duration commencing at "Motor Switch-On" initialisation until the model has passed the competition required altitude. The ALMRT shall determine the 'Motor-Stop Height' in accordance with the following procedure:
- (a) Immediately on application of power the status of the ALMRT shall indicate in a manner that is clearly visible by the competitor that it is correctly reset and armed to operate.
 - (b) Initialisation following power-up, and the acquisition of the ground level (zero height reference) reading should be accomplished with due allowance for any time required for the pressure sensor device to stabilise.
 - (c) This function shall be carried out with the model at ground level at the location specified within the competition rules.
 - (d) A maximum of 2 seconds is permitted for the zero-height reference level to be determined, during which period the ALMRT shall not respond to any Motor-Start signal instituted by the competitor or resulting from receiver start up procedures.
Note: A Motor-Start command is defined as any control signal exceeding 1.2 milliseconds. Before the motor-start command the ALMRT shall not output a control signal greater than 0.9 milliseconds which means motor-off.
 - (e) The 'Start Height' determination and the start of the Motor Run Timer will be initiated by the Motor-Start signal instituted by the competitor.
- 2.6 Motor Run Timer
- 2.6.1 The Motor Run Timer function must operate with the accuracy specified in item 2.3.5.
- a) On initial power-up of the ALMRT, the Motor Run Timer must be protected from false triggering during the period defined in Item 2.5 (d). After this period, the Motor Run Timer shall commence operation and 'Motor Off' timer shall be initiated on the first occasion that the command signal exceeds 1.2 milliseconds.
 - b) After the Motor Run Timer has been triggered, the timing of the motor run ceases when either the command signal reduces below 1.2 milliseconds, or the elapsed time reaches 30 seconds and a

definite stop command must be applied to the ESC by the AMRT.

- c) Once the definite stop command has been applied to the ESC, the system must not respond to any further changes in the motor command signal.

2.6.2 A small amount of hysteresis is permitted in the sensing of the 1.2 milliseconds motor command signal to eliminate malfunction with jittering command signals.

2.7 ESC Compatibility

There is no defined standard for the manner in which ESCs (electronic speed controllers) initialise to the range of motor command signals. In addition, programmable transmitters allow the user to set the motor command signals to any value within the programmable range.

2.7.1 Commercial pressures will encourage ALMRT manufacturers to configure their products to operate with the majority of available ESC devices.

2.7.2 The F5L specification of motor-on threshold at 1.2 milliseconds has been selected as being compatible with most known equipment but cannot be a guarantee of correct operation for every combination of transmitter, receiver and ESC.

3.0 COMPETITION PROCEDURES

3.1 Installation Environment

3.1.1 The installation and functioning of the ALMRT, the transmitter and its programming, receiver and ESC, is the sole responsibility of the competitor, who must ensure that they operate together in the correct manner as required for the rules of the competition.

3.1.2 The ALMRT must be installed in such a way that it is protected from pressure fluctuations other than changes in atmospheric pressure which result from the height of the model above ground level.

3.1.3 The ALMRT must be placed inside the model in the state as supplied by the manufacturer. The use of any method that modifies the true barometric pressure at any time is prohibited.

3.1.4 Adequate venting shall be provided to minimize any uneven pressure distribution within the model in the region where the ALMRT is installed.

3.1.5 The ESC must always operate via its series connection to the ALMRT and not with a direct connection to the receiver.

4.0 VERIFICATION OF ALMRT's

This section details procedures for verifying the operation of Approved ALMRTs for use in the F5L competition environment. It is provided as a guide for both competitors and competition organizers.

4.1 Purpose of Verification

All ALMRTs used in an F5L competition are required to be of an Approved Type. The verification process serves to ensure that no individual AMRT operates in a manner different from others in use at the competition. As such, comparison methods are employed for 'Start Height' verification, in place of a requirement to undertake absolute accuracy measurements.

4.2 Verification Procedures

4.2.1 Firmware Revision Level

An ALMRT with connected display, will show the firmware revision level for three seconds immediately after the application of power. Otherwise, the firmware revision level is printed or labeled on the device.

4.2.2 Operational Sequence

For this procedure, the ALMRT has to be supplied with power, a command signal input and a means of monitoring the motor "run/stop" signal that is supplied to the ESC.

Whilst this procedure is applicable to a complete aircraft installation, for reasons of safety and when verifying an ALMRT, it should be removed from the aircraft and tested independently.

The procedure is as follows:

- a) With the command signal at "Motor-Off", apply power to the ALMRT. After the optional initial display of the firmware revision level, the ESC signal will be in the "Motor-Stop" state.
- b) Apply a "Motor-On" command.
- c) During the first 30 seconds the ESC signal follows the command signal.
- d) 30 seconds after the "Motor-On" command, the ESC signal will change to "motor-stop".
- e) Apply a "Motor-Off" command. Any further "Motor-On" or "Motor-Off" commands should not take any action and leave the EDC signal in the "Motor-Stop" state.

- f) Switch power, apply a “Motor-On” command and raise the ALMRT from ground above 90m. The ESC signal will go in the “Motor-Stop”. Any further “Motor-On” or “Motor-Off” commands should not take any action and leave the EDC signal in the “Motor-Stop” state even when varying height.
- g) The “Motor-Stop” state must be established either by exceeding 30s or 90m height whichever comes first.

4.2.3 ‘Start Height’ Detection

Verification of the measured ‘Start Height’ value requires the use of a ‘vacuum’ chamber. Such a chamber comprises an airtight container from which air can be extracted to create a pressure reduction. Air equivalent to approximately 3% of the chamber capacity has to be extracted in 15 to 20 seconds in order to simulate a typical model climb. On completion of the climb simulation, a slight increase in pressure is required in order to emulate the transition to gliding flight.

For verification purposes it is only necessary to perform tests starting at local ambient pressure. Slight air leakage of the chamber is acceptable and aids the gliding flight simulation.

Within the chamber, facilities are required to provide power and simultaneous command signals to a number of AMRTs.

A known, good AMRT (preferably with a certificated calibration) is used as the reference standard against which other AMRTs are compared.

The procedure is as follows.

- a) Place the device(s) to be tested in the chamber together with the reference device.
- b) Readout display(s) may be attached either before or after the pressure change has been applied.
- c) Close the chamber and allow a short period for the pressure to stabilise. (Some variation may occur due to temperature differences introduced by handling).
- d) With the Motor Command set to “motor off”, apply power to the AMRTs. This causes the zero height reference to be detected by the AMRT(s).
- e) Apply a Motor Run command.
- f) Reduce the chamber pressure at an appropriate rate to simulate a climb to approximately 150 metres.
- g) Apply a Motor Stop command and immediately allow the pressure to begin increasing to simulate a descent of gliding flight.
- h) Wait for a minimum of 10 seconds.
- i) Observe ‘Start Height’ readings for all AMRTs.
- j) Analyze the results by comparison with the known, good AMRT.

Any AMRT that deviates by more than plus or minus 1.25 metre from the monitoring AMRT should be subject to further investigation.

Note: The overall tolerance for absolute accuracy specified in Section 2 is the worst case figure when all possible discrepancies are taken into consideration.

(The 1.25 metre figure may be subject to revision when more experience has been gained in the competition environment).

Repeat the procedure simulating other climb heights as considered necessary.

5.0 GUIDANCE FOR COMPETITION ORGANISERS ON THE USE OF MONITORING AMRTS

Competition rules allow competition organizers to carry out random checks on competitors’ equipment at any time during a competition. One aspect of this could be the use of a monitoring ALMRT to be fitted in a competitor’s aircraft for a competition flight. Item 4.1.9 defines that the appropriate connectors in an aircraft must be accessible.

5.1 Use of Monitoring AMRT in Competitors’ Aircraft

- 5.1.1 A monitoring AMRT with logging function as used in F5J competitions may be fitted at the command signal input of an AMRT by means of a Y connection lead. This permits the monitoring AMRT to record the motor command signals and height measurement of a flight.

The rating and quality of the Y lead must be such that it does not degrade the performance of the competitor’s own installation.

- 5.1.2 The ‘Motor-Off Height’, as determined by the monitoring AMRT should be in close (less than 0.1s) agreement with the reading taken by the competitor’s own ALMRT.

Allowing for variations in pressure distribution within the aircraft, agreement within plus or minus

2.0 metre is considered acceptable.

Note: The 2.0 metre figure may be subject to revision when more experience has been gained in the competition environment.

- 5.1.3 The “Motor-Off Time”, as determined by the monitoring AMRT should be in close agreement with the reading taken by the competitor’s own ALMRT.

6.0 APPROVAL PROCEDURES

6.1 Check list of features to be verified when evaluating compliance

- (a) Display of firmware revision level.
- (b) Operational sequence (as in 5.2.2).
- (c) Accuracy of Motor On pulse sensing.
- (d) Accuracy of timing of motor run.
- (e) Accuracy of 30 second period for motor cut.
- (f) Accuracy of ISA calculation across full ambient pressure range.
- (g) Accuracy of “Motor-Off Height” determination.
- (h) Dynamic response of “Motor-Off Height” determination.
- (i) Effect of temperature variations on “Motor-Off Height” accuracy.
- (j) Prevention of motor restart.
- (k) Verification that any additional features not required by the specification (eg buttons, display, flight logging) do not affect the essential operation of the device.

6.2 Submission of ALMRT’s for Approval

The CIAM EDIC Working Group is responsible for the approval process. The services of a technical expert may be used to undertake the practical testing on behalf of the Working Group.

Further details will be advised in due course, together with the specific requirements concerning the presentation of devices for approval.

Devices submitted for approval must be to normal production standard and must incorporate firmware at the revision level that is to be evaluated.

Supporting documentation shall include the sensor manufacturer’s data sheet on the pressure sensor device with particular reference to long term drift and temperature variation characteristics.

Additional supporting information may be required at the request of the technical expert undertaking the evaluation on behalf of the CIAM EDIC Working Group.

Approval, when granted, will relate to a specific hardware/firmware combination. Any subsequent modification to hardware or firmware must be notified to the CIAM committee and advice will be provided concerning any requirements for upgrading the previously granted approval.

6.3 Withdrawal of Approved Status

An approved ALMRT may have its Approved Status withdrawn if inconsistencies of performance are found in further examples of the ALMRT.

If, subsequent to the granting of an approval status, the rules of the competition are amended in a manner that affects the technical specification of the ALMRT, the validity of all ALMRT’s on the Approved List will be subject to review.

EDIC WORKING GROUP TERMS OF REFERENCE

1.0 Background

- (a) Electronic devices are already in use in aeromodelling or space modelling competitions for scoring purposes. Devices need to be manufactured under specific technical specifications and then need to be checked whether they meet those specifications and finally be approved for official use. During the December 2013 meeting, CIAM Bureau decided to form a Working Group which will be responsible for this task.
- (b) All projects undertaken by the EDIC WG are at the request of CIAM Bureau or a CIAM Subcommittee Chairman.

2.0 Mission

The mission of this CIAM Working Group is:

- (a) To issue technical specifications for all electronic devices used for CIAM competitions.
- (b) To communicate with companies which are willing to manufacture such electronic devices.
- (c) To define the approval procedure and the necessary steps in order to evaluate such devices.
- (d) To test the devices according to the procedures defined.
- (e) To issue and maintain an approval list of all such devices.
- (f) To provide advice, to CIAM, on any matter relating to electronic devices that are used or can be used for aeromodelling or space modelling competitions.

3.0 Membership

The working group will comprise 3 (three) appointed members:

- (a) One person who will be the Chairman and will be appointed by CIAM Bureau. He will be responsible for all CIAM running projects.
- (b) Two persons per project who will be appointed by the relevant CIAM Subcommittee.

4.0 Term

The Working Group Chairman shall be appointed for a two-year term and may be re-appointed.

The two persons appointed by the relevant CIAM Subcommittee, will be members of this Working Group for the life of the specific project.

5.0 Transaction of business

- (a) The Working Group may plan face-to-face meetings in conjunction with other CIAM activities at which all, or the majority of, Working Group members are present.
- (b) The normal business of the Working Group is expected to be conducted through the mediums of email and VoIP (Voice over IP) services.
- (c) Working Group members are not entitled to claim any casual expenses or travel costs from the CIAM unless these are expressly pre-approved by CIAM Bureau.

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FAI AUTHORITY

The competition will be conducted under the authority granted by the FAI, according to the regulations of the Sporting Code of the FAI, General Section, and Section 4 as approved by the CIAM and validated by the FAI, and these rules. All participants accept these rules and the FAI regulations as binding by registering in the competition.

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